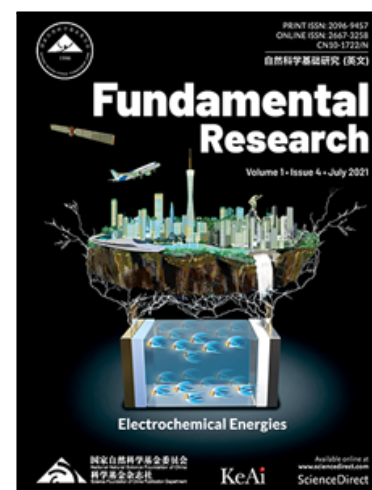


Journal Pre-proof

Knowledge Graph Quality Control: A Survey

Xiangyu Wang, Lyuzhou Chen, Taiyu Ban, Muhammad Usman,
Yifeng Guan, Shikang Liu, Tianhao Wu, Huanhuan Chen

PII: S2667-3258(21)00165-5
DOI: <https://doi.org/10.1016/j.fmre.2021.09.003>
Reference: FMRE 107



To appear in: *Fundamental Research*

Received date: 9 May 2021
Revised date: 12 August 2021
Accepted date: 17 August 2021

Please cite this article as: Xiangyu Wang, Lyuzhou Chen, Taiyu Ban, Muhammad Usman, Yifeng Guan, Shikang Liu, Tianhao Wu, Huanhuan Chen, Knowledge Graph Quality Control: A Survey, *Fundamental Research* (2021), doi: <https://doi.org/10.1016/j.fmre.2021.09.003>

This is a PDF file of an article that has undergone enhancements after acceptance, such as the addition of a cover page and metadata, and formatting for readability, but it is not yet the definitive version of record. This version will undergo additional copyediting, typesetting and review before it is published in its final form, but we are providing this version to give early visibility of the article. Please note that, during the production process, errors may be discovered which could affect the content, and all legal disclaimers that apply to the journal pertain.

© 2021 The Authors. Publishing Services by Elsevier B.V. on behalf of KeAi Communications Co. Ltd.

This is an open access article under the CC BY-NC-ND license
(<http://creativecommons.org/licenses/by-nc-nd/4.0/>)

Knowledge Graph Quality Control: A Survey

Xiangyu Wang, Lyuzhou Chen, Taiyu Ban, Muhammad Usman, Yifeng Guan, Shikang Liu, Tianhao Wu
Huanhuan Chen*

University of Science and Technology of China

96, JinZhai Road, Hefei, China, 230026

Abstract

A knowledge graph (KG), a special form of semantic network, integrates fragmentary data into a graph to support knowledge processing and reasoning. KG quality control is important to the utility of KGs. It is essential to investigate KG quality and the parameters influencing KG quality to better understand its quality control. Although many works have been conducted to evaluate the dimensions of KG quality, quality control of the construction process, and enhancement methods for quality, a comprehensive literature review has not been presented on this topic. This paper intends to fill this research gap by presenting a comprehensive survey on the quality control of KGs. First, this paper defines six main evaluation dimensions of KG quality and investigates their correlations and differences. Second, quality control treatments during KG construction are introduced from the perspective of these dimensions of KG quality. Third, the quality enhancement of a constructed KG is described from various dimensions. This paper ultimately aims to promote the research and applications of KGs.

Keywords: Knowledge graph, Quality control, Quality evaluation, Quality enhancement

1. Introduction

The last decade has witnessed a surge of applications and research on knowledge graph (KGs). A KG is generally defined as “a multi-relational graph composed of entities as nodes and relations as different types of edges” [1]. Despite various specific definitions of a KG, it is often referred to as a knowledge network extracted from the real world [2–4]. Moreover, Ehrlinger *et al.* [5] proposed a new definition of a KG based on ontologies and reasoners deriving new knowledge, leading to a definition of a KG with respect to the KG construction process. From the perspective of applications and quality control, a KG is a special form of semantic network that integrates triplets into a graph to support knowledge processing and reasoning. Compared with traditional knowledge representation methods, a KG effectively and intuitively expresses the relation between entities [6]. Therefore, KGs are widely applied in various fields, such as semantic search [7, 8], question and answering systems [9], personalized recommendation systems [10, 11], and decision support systems [12, 13]. The quality of a KG determines its utility in specific application scenarios. In this context, KG quality control is essential to the practical application of KGs.

The evaluation of KG quality¹ is essential for the quality control of KGs. There are six main dimensions of KG

quality: accuracy, consistency, completeness, timeliness, trustworthiness, and availability [4, 14, 15], as shown in Figure 1. Each evaluation dimension of KG quality can be utilized as an indicator of certain aspects of a KG [16], providing an objective standard for KG quality control. Moreover, the evaluation dimensions of KG quality are not independent of each other. Understanding the correlation between these dimensions is helpful in choosing an appropriate method to control KG quality and meet the specific requirements² linked with KG quality in different applications.

For KG construction, a number of highly efficient methods for various processes³ of KG construction have been proposed [17]. However, these methods inevitably introduce factors that can negatively influence KG quality. Consequently, the quality control of each KG construction process is very important to KG quality [18–20]. It is necessary to figure out the corresponding quality dimensions that are affected by each process.

Besides KG construction, the enhancement of KG quality is also a crucial step in KG quality control. In general,

²In some KG-based applications, there may be specialized requirements for different evaluation dimensions of KG quality. For example, KG-based information retrieval requires a KG with high trustworthiness and accuracy, while availability is not a major concern.

³KG construction consists of a series of procedures, including knowledge extraction and knowledge fusion, and there are corresponding processes in each procedure, such as named entity recognition and entity relation extraction in knowledge extraction.

*Corresponding author: Huanhuan Chen (hchen@ustc.edu.cn)

¹“The evaluation of KG quality” refers to obtaining metrics on various aspects of a KG.

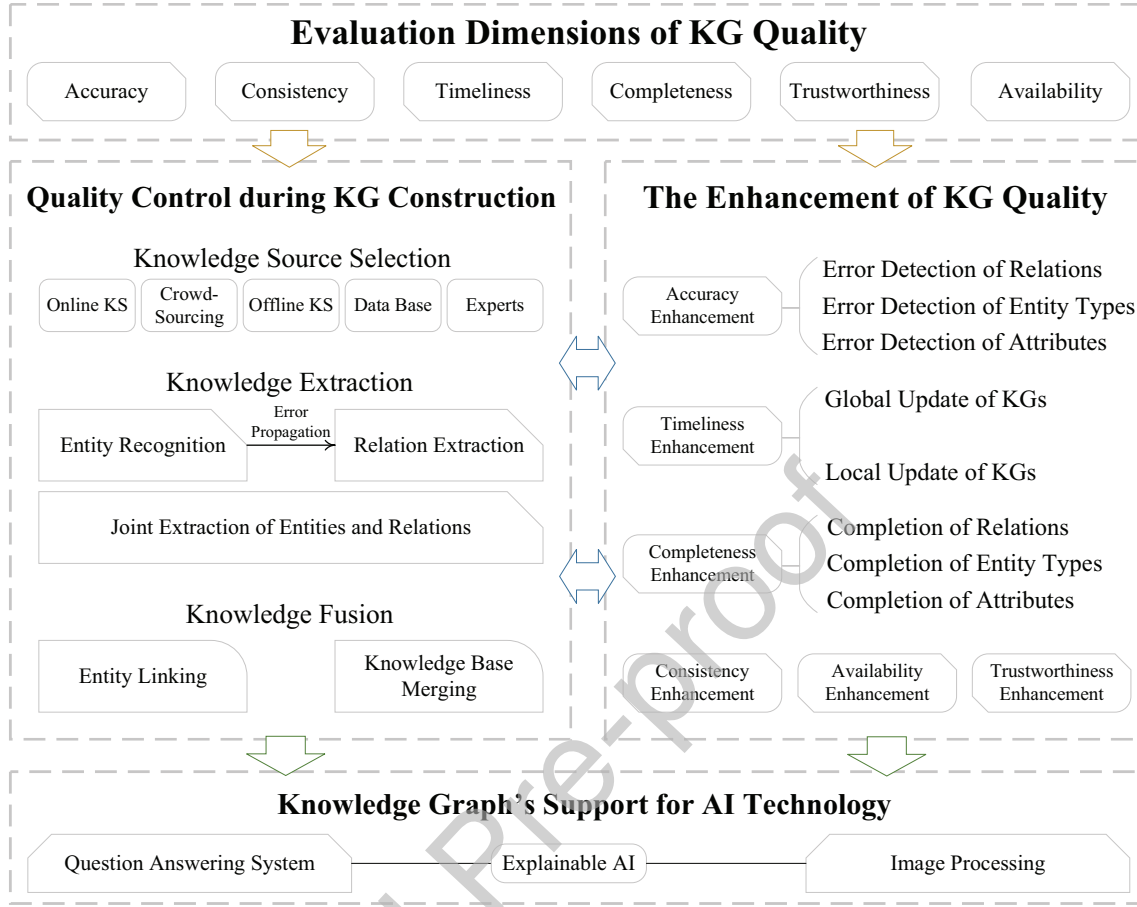


Figure 1: Organization of this paper

KGs need to be maintained [21] and updated [22] to meet new requirements and remain up-to-date. Besides, errors will inevitably still exist in the constructed KG, which need to be corrected [23] to achieve KG quality control. Therefore, to meet new application requirements and solve the problems that remain from KG construction, it is necessary to enhance the quality of the constructed KG.

Although a significant amount of work has been conducted to evaluate the dimensions of KG quality, quality control of the KG construction process, and enhancement methods for quality, a comprehensive literature review has not been presented on this topic [2, 17]. This work intends to fill this research gap. Specifically, this paper makes the following contributions:

- To provide guidance for KG quality control, this paper presents a comprehensive introduction to the evaluation dimensions of KG quality and analyzes the correlation between them.
- This paper analyzes the influences of various processes on KG quality and summarizes the corresponding methods for quality control during KG construction.

tion.

- This paper summarizes the main enhancement methods of KG quality and analyzes the correlation between these methods and corresponding dimensions of KG quality.

The remainder of this paper is organized as follows. The evaluation dimensions of KG quality are introduced in Section 2. Quality control during KG construction is introduced in Section 3. The enhancement of KG quality is introduced in Section 4. In Section 5, KG support for AI methods is discussed. Finally, we conclude the paper in Section 6.

2. Evaluation Dimensions of KG Quality

2.1. Related Framework and KG Quality Dimension Analysis

There are some well-known evaluation frameworks for data and information quality evaluation, such as Zaveri *et al.*'s framework [25] and Stvila *et al.*'s framework [14].

Table 1: Definitions of evaluation dimensions proposed in related frameworks

Dimension	Sub-dimension	Definition or Description	Relationship with our framework
Accessibility [24]	Availability	The extent to which data (or some portion of it) are present, obtainable, and ready for use	Availability
	Interlinking	The degree to which entities that represent the same concept are linked to each other	
	Performance	The efficiency of a system that binds to a large dataset	
Intrinsic [24]	Semantic accuracy	The degree to which data values correctly represent the real-world facts	Accuracy
	Consistency	Consistency means that a knowledge base is free of contradictions with respect to particular knowledge representation and inference mechanisms	Consistency
	Completeness	The degree to which all required information is present in a particular dataset	Completeness
Contextual [24]	Relevancy	The provision of information in accordance with the task at hand and importance to a user's query	Accuracy, Completeness
	Trustworthiness	The degree to which the information is accepted to be correct, true, real, and credible	Trustworthiness
	Timeliness	How up-to-date data are relative to a specific task	Timeliness
Representation [24]	Interoperability	The degree to which the format and structure of the information conforms to previously returned information as well as data from other sources	Interoperability
	Versatility	The availability of the data in different representations and in an internationalized manner	Availability
Intrinsic [25]	Semantic consistency	The extent of consistency in using the same values and elements to convey the same concepts and meanings in an information object	Consistency
	Structural consistency	The extent to which similar attributes or elements of an information object are consistently represented using the same structure, format, and precision	
	Precision, Completeness	The granularity or precision of the model or content values of an information object according to some general-purpose IS-A ontology, such as WordNet	
Relational contextual [25]	Currency	The age of an information object	Timeliness
	Verifiability	The extent to which the correctness of information is verifiable or provable in the context of a particular activity	Accuracy, Trustworthiness
	Accuracy	The degree to which an information object correctly represents some other information, contextual object, process, or phenomenon in the context of a particular activity or culture	Accuracy
	Accessibility	Speed and ease of locating and obtaining an information object relative to a particular activity	Availability
	Relevance	The extent to which information is applicable in a given activity	Accuracy, Completeness
	Informativeness, Redundancy	The extent to which the information is new or informative in the context of a particular activity or community	Timeliness, Completeness, Accuracy
Reputation [25]	Authority	The degree of reputation of an information object in a given community or culture	Trustworthiness
Completeness[26]	Completeness of mandatory attributes	The attributes that are mandatory for a complete representation of a real-world entity must contain values and cannot be null	Completeness
	Completeness of optional attributes	Optional attributes should not contain invalid null values	
Accuracy [26]	Precision	Attribute values should be accurate according to linguistics and granularity	Accuracy
Consistency [26]	Semantic consistency	Data are semantically consistent	Consistency
	Value consistency	Data values are consistent and do not provide conflicting or heterogeneous instances	
	Format consistency	Data formats are consistently used	
Validity [26]	Business rule compliance	Data should comply with business rules	Availability
	Meta-data compliance	Data should comply with its meta-data	

The definitions of some classic quality dimensions of data and linked data are listed in Table 1.

As a special form of data and information, the quality evaluation dimensions of a KG have overlapping and similar definitions to some evaluation dimensions in the above-mentioned evaluation frameworks. However, due to the different requirements of the application tasks and data structure, some dimensions are irrelevant or redun-

dant to KGs, with the exception of some rare cases. For example, some researches classify interoperability as a sub-dimension that assesses the extent of knowledge that remains in the same format. However, this is secondary to KGs because many existing KGs contain multiple modal data. Some minor evaluation dimensions are listed in Table 2.

This paper aims to propose a universal evaluation frame-

work using generalized evaluation dimensions to cover most of the evaluation perspectives of a KG. By summarizing and analyzing the dimensions of the existing classic framework, this paper selects the main dimensions that are closely related to KGs based on of the above framework. Subsequently, this paper summarizes and integrates these into some of the most important dimensions, including accuracy, completeness, consistency, timeliness, trustworthiness, and availability. Table 1 lists the evaluation dimensions from related frameworks and their relationship with our framework.

2.2. Dimensions of KG Quality

The evaluation of KG quality is multi-dimensional, and the six evaluation dimensions for KG quality have been introduced above: accuracy, completeness, consistency, timeliness, trustworthiness, and availability. Although the relevant frameworks have definitions for these dimensions, it is necessary to introduce them specifically here in the context of KGs. This paper investigates the definitions and corresponding formulas of the above dimensions in related works for the completeness of the research, as listed in Table 3.

2.2.1. KG Accuracy

KG accuracy refers to the degree of correctness of facts in a KG [24]. KG accuracy has often been discussed in other works. According to Wang *et al.*, KG accuracy is defined as “the extent to which knowledge are correct, reliable, and certified free of error” [15]. The work in [28] treated the accuracy of knowledge as a synonym for knowledge quality. Generally, KG accuracy is evaluated using gold standards⁴ and expert verification. Additionally, some specialized methods have been designed to evaluate KG accuracy in KG-based tasks. Erber *et al.* [27] evaluated KG accuracy using the functional dependency between the attributes of entities. Lei *et al.* [33] evaluated KG accuracy through the ratios of inaccurate labels, annotations, and classifications of entities. KGs are widely used in various applications, such as KG-based information retrieval and question and answering systems. Because incorrect facts often lead to incorrect knowledge, accuracy is a prolific evaluation measure for KG-based applications.

2.2.2. KG Consistency

KG consistency is defined as the degree to which the knowledge of a KG does not contradict itself [14]. For example, if a KG contains the following triples: (*James Smith*, *IsFatherOf*, *Jack Smith*), (*James Smith*, *IsHusbandOf*, *Marry Smith*), and (*Marry Smith*, *IsSisterOf*, *Jack Smith*). The inferred knowledge “*Marry Smith is the mother of Jack Smith*” contradicts the semantic expression of the

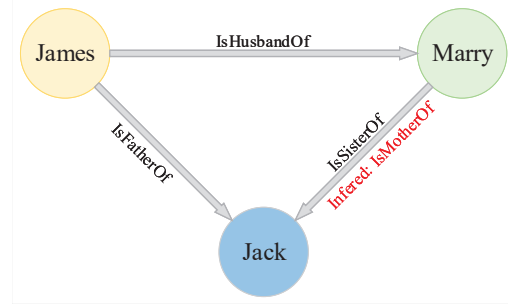


Figure 2: Example of inconsistent knowledge

third triple, making the KG inconsistent, as shown in Figure 2. Contrary to KG accuracy, KG consistency focuses on the logical correctness of a KG rather than the semantic correctness of triples. For example, the knowledge from a KG may no longer conform to the facts over time, but it has no contradictions. In this case, the KG can hold high consistency even with low accuracy. On the contrary, good accuracy generally leads to good consistency because existing facts are usually self-consistent.

Researchers have proposed various methods for evaluating KG consistency. Hogan *et al.* [29] used the proportion of entities that belong to disjoint classes as the assessment criteria for KG consistency. Zaveri *et al.* [34] evaluated KG consistency by generating a set of manually-verified schema axioms. Furthermore, methods such as consistency-based error detection [35] lead to an improvement in KG accuracy. Moreover, many KGs check consistency through knowledge reasoning before release, aiming to discover and correct incorrect knowledge, thereby leading to improved KG accuracy [2].

2.2.3. KG Completeness

KG completeness refers to the extent to which a KG covers the knowledge of interest [25]. Knowledge of interest is a relative concept, so KG completeness is also a relative quality dimension. Generally, a KG is considered to contain good completeness if it meets the knowledge interests of the underlying applications. For example, Google Knowledge Graph includes a large number of common senses and has high completeness for general searches. However, Google Knowledge Graph may be incomplete for some specific professional domains. Compared with KG accuracy, KG completeness does not have strict requirements for the correctness of knowledge. It focuses on whether the relation between entities is explicitly expressed and whether the entity attributes are completed. Different methods have been proposed to evaluate KG completeness. For instance, Mendes *et al.* [36] used the number of classes, entities, attributes, etc., to define related indicators and scoring functions to evaluate KG completeness, while Luggen *et al.* [37] used nonparametric statistical techniques to estimate completeness. KG completeness is crucial for KG-based applications; there-

⁴In this paper, gold standard refers to a some reference dataset that can be used for comparison of knowledge.

Table 2: Irrelevant or redundant dimensions of a KG in related frameworks compared to our framework

Dimension	Sub-dimension	Definition
Representation	Interoperability [24]	Interoperability is the degree to which the format and structure of the information conforms to previously returned information as well as data from other sources.
Intrinsic	Conciseness [24]	Conciseness refers to the minimization of redundancy of entities at the schema and data levels.
Intrinsic	Syntactic validity [24]	Syntactic validity is defined as the degree to which an RDF document conforms to the specification of the serialization format
Consistency	Non-redundancy [26]	The data are recorded in exactly one place
Consistency	Referential integrity [26]	Data relationships are represented through referential integrity rules
Relational contextual	Security [25]	The extent to which information is protected from harm in the context of a particular activity
Relational contextual	Naturalness [25]	The degree to which the model and content of an information object are semantically close to the objects, states, or processes they represent in the context of a particular activity

fore, recent studies have focused on the development of techniques for improving the completeness of KGs.

2.2.4. KG Timeliness

KG timeliness is defined as the degree to which knowledge is up-to-date [38]. For example, a daily updated KG can be considered to have high timeliness. The demand for KG' timeliness depends on the specific application scenarios, similar to KG completeness. For example, for a KG of common sense, KG timeliness could reach the requirement of applications if it has been updated in recent years. A KG modeled for research on COVID-19 must be maintained with real-time updates. Interestingly, there exists a relationship between KG timeliness and accuracy. For example, the low timeliness triple (Trump, IsPresidentOf the, United States) could be inaccurate in most situations. Naumann *et al.* [28] defined KG timeliness as the time range since the last update or the last verification of the statement, and Frber *et al.* [38] used this method to assess timeliness. Another study defined KG timeliness as the ratio of time from the last update to the time from the first existence of knowledge [14].

2.2.5. KG Trustworthiness

KG trustworthiness is the degree of objectivity, authority, and verifiability of a KG. It describes how credible the knowledge in a KG is accepted to be [39]. Trustworthiness is a comprehensive dimension with a subjective perspective, such as authority of knowledge [30]; thus, it has a strong relationship with other dimensions. For instance, improvements in other measures, such as accuracy, timeliness, and consistency of a KG, increase knowledge

authority, leading to an improvement in KG trustworthiness. Moreover, KG trustworthiness can be affected by knowledge source selection and KG construction methods [15, 28]. Generally, when the source of the KG is credible and its construction method has superior performance, the reliability of the constructed KG is high.

KG trustworthiness can be assessed through the trustworthiness of knowledge sources. Gamble [40] constructed a decision network of knowledge source graphs for trustworthiness evaluation. Furthermore, trustworthiness can also be evaluated by assessing KG accuracy because high KG accuracy usually implies high trustworthiness. In addition to these perspectives, trustworthiness can also be considered as an a priori impression of an unverified KG [31], making trustworthiness an empirical metric.

2.2.6. KG Availability

KG availability is the extent to which knowledge is convenient to use. In other words, it refers to how easily and quickly the knowledge of KGs can be retrieved [15]. However, it is difficult to quantitatively evaluate KG availability. We divide it into the following three parts: 1) For machines, KG availability measures whether a KG is in a machine-readable form or can be downloaded. 2) For users, KG availability refers to the ability to support different languages, usage of technical queries with, e.g., SQL, etc. 3) For fast retrieval of knowledge, KG availability refers to the speed of data response, degree of interconnection between entities with similar properties, and form of knowledge representation [14, 29, 31, 38, 41]. In addition, the evaluation and improvement methods of KG availability [32, 42] differ from other evaluation dimensions. To enhance KG availability, it is necessary to start with the

Table 3: Definitions and formulas of six dimensions proposed in related research

Dimension	Definition	Formula	Reference
Accuracy	Syntactic validity (syntactic accuracy) defines the degree to which an RDF document conforms to the specification of the serialization format.		Zaveri et al. [14]
	Semantic accuracy regards whether the data value represents the correct state of an object.	$1 - \frac{\text{inaccurate instances}}{\text{total no. of instances}} + \frac{\text{balanced distance metric}}{\text{total no. of instances}}$	
	Data are certified error-free, accurate, correct, flawless, reliable, errors can be easily identified, data integrity is ensured, precise.		Wang et al. [15]
	The accuracy dimension describes the proximity of data value representations of an object to their real-world states.		Fürber et al. [27]
	Accuracy is the ratio of the number of correct values in a source to the total number of values in the source.	$\frac{\text{no. of correct values in a source}}{\text{total no. of values in the source}}$	Naumann et al. [28]
Consistency	Consistency means that a knowledge base is free of (logical/formal) contradictions with respect to particular knowledge representation and inference mechanisms.	$\frac{\text{no. of entities described as members of disjoint classes}}{\text{total no. of entities described in the dataset}}$	Zaveri et al. [14]
	Consistency defines no contradictions in the data.		Hogan et al. [29]
Completeness	Completeness refers to the degree to which all required information is present in a particular dataset.	Population Completeness: $\frac{\text{no. of values represented for a specific property}}{\text{total no. of values for a specific property}}$ Population Completeness: $\frac{\text{no. of real-world objects are represented}}{\text{total no. of real-world objects}}$ Interlinking Completeness: $\frac{\text{no. of instances in the dataset that are interlinked}}{\text{total no. of instances in a dataset}}$	Zaveri et al. [14]
	Breadth, depth, and scope of information contained in the data.		Wang et al. [15]
Timeliness	Timeliness is the average age of the data in a source.	$\frac{\sum \text{the data in a source the age of data}}{\text{no. of the data in a source}}$	Naumann et al. [28]
	Timeliness measures how up-to-date data are relative to a specific task.	$1 - \frac{\text{observation time} - \text{last modified time}}{\text{ideal freshness}}$	Zaveri et al. [24]
Trustworthiness	Trustworthiness defines the degree to which the information is accepted to be correct, true, real, and credible.		Zaveri et al. [14]
	Trustworthiness is the degree to which data are accepted as correct by the user.	$E(\frac{\text{no. of correct values in a source}}{\text{total no. of values in the source}})$	Naumann et al. [28]
	Trustworthiness represents the reputation of an entity or dataset either as a result of direct experience or from the recommendations of others.		Gil et al. [30]
Availability	Availability is the extent to which information is available or easily and quickly retrievable.		Bizer et al. [31]
	Availability refers to the proper functioning of all access methods.		Flemming et al. [32]
	Availability of a dataset is the extent to which information (or some portion of it) is present, obtainable, and ready for use.	$\frac{\text{no. of times the server responded to SPARQL requests}}{\text{total no. of SPARQL requests}}$ $1 - \frac{\text{no. of accessing fails}}{\text{total no. of HTTP-GET requests}}$	Zaveri et al. [24]

design, architecture, and other aspects of the KG.

2.3. Correlation between the Dimensions

Because the improvement of one dimension of a KG is always based on updating of the knowledge therein and often leads to the improvement of other dimensions, the dimensions are not independent of each other. The correlation of the six main dimensions of KG quality was partially discussed above. To clarify these associations, they are enumerated in Table 5 based on the definition in our framework and the discussion of related classic frameworks.

3. Quality Control during KG Construction

Each process during KG construction influences KG quality. Therefore, KG quality control can be divided into different corresponding periods. For example, the process of knowledge source selection determines the quality of knowledge, i.e., its credibility and accuracy. Moreover, the processes of knowledge extraction and knowledge fusion determine the accuracy and recall of the extracted knowledge elements, which affect the KG quality dimensions of accuracy and completeness. First, this section introduces quality control during knowledge source selection and lists methods for selecting high-quality knowledge sources. Second, this section discusses the influence of knowledge extraction on KG quality. Third, knowledge

Table 4: References to dimensions of KG quality

Dimension	Reference	
Accuracy	Wang <i>et al.</i> [15]	Furber <i>et al.</i> [27]
	Naumann <i>et al.</i> [28]	Lei <i>et al.</i> [33]
Consistency	Hogan <i>et al.</i> [29]	Li <i>et al.</i> [35]
	Zaveri <i>et al.</i> [34]	Paulheim <i>et al.</i> [2]
Completeness	Mendes <i>et al.</i> [36]	
	Luggen <i>et al.</i> [37]	
Timeliness	Naumann <i>et al.</i> [28]	Zaveri <i>et al.</i> [14]
	Farber <i>et al.</i> [38]	
Trustworthiness	Gil <i>et al.</i> [30]	Wang <i>et al.</i> [15]
	Naumann <i>et al.</i> [28]	Gamble <i>et al.</i> [40]
	Bizer <i>et al.</i> [31]	
Availability	Wang <i>et al.</i> [15]	Zaveri <i>et al.</i> [14]
	Farber <i>et al.</i> [38]	Bizer <i>et al.</i> [31]
	Bizer <i>et al.</i> [41]	Hogan <i>et al.</i> [29]
	Flemming <i>et al.</i> [32]	Gueret <i>et al.</i> [42]

Table 5: References to dimensions of KG quality

	Accuracy	Completeness	Consistency	Timeliness	Trustworthiness	Availability
Accuracy		↓ [14]	↑ [14]	↑ [43]	↑ [44]	↓ [45]
Completeness	↓ [14]		— [14]	↓ [43]	↑ [15]	↑ [46]
Consistency	↑ [14]	— [14]		↓ [43]	↑ [15]	— [46]
Timeliness	↑ [43]	↓ [43]	↓ [43]		↑ [14]	— [14]
Trustworthiness	↑ [44]	↑ [15]	↑ [15]	↑ [14]		— [14]
Availability	↓ [45]	↑ [46]	— [46]	— [14]	— [14]	

*‘↑’, ‘↓’, and ‘—’ indicate that the enhancement of the two dimensions is positively, negatively, or almost entirely not correlated, respectively

fusion is introduced for further KG quality control. The schema of this section is illustrated in Figure 3.

3.1. Knowledge Source Selection

The selection of knowledge sources⁵ has a significant impact on KG quality and essentially determines the upper limit of KG quality. The quality of knowledge sources can be evaluated generally from the credibility and relevance perspectives. Credibility refers to the degree to which knowledge sources are convincing, and relevance refers to the degree to which knowledge sources are related to the knowledge of interest. The existing methods of knowledge source selection can mainly be categorized as 1) designing

credibility evaluation methods of knowledge sources to select high-credibility knowledge sources or 2) improving the relevance of knowledge sources through noise filtering.

Many knowledge sources contain multi-modal knowledge. However, most of the knowledge source selection methods discussed in this paper are based on natural language processing (NLP) for generality of the research. Moreover, the methods of knowledge presentation of knowledge sources are diverse. For example, knowledge can be acquired from the web, crowdsourcing data, offline text, databases, and experts. These different knowledge sources have different characteristics, and the corresponding knowledge source selection methods are different. The schema of knowledge source selection is illustrated in Figure 4.

3.1.1. Online Knowledge Source Selection

KG construction commonly uses online knowledge sources, which are generally accessible to the public. Online knowledge sources are mostly obtained by accessing online KGs

⁵In the construction of a KG, different processing methods are used according to the knowledge source medium, e.g., picture, video, and audio. Due to the diversity of these methods in different fields, summarization is cumbersome. However, natural language processing (NLP) is a more common method of KG construction. Therefore, the scope of knowledge sources discussed in this paper is limited to NLP, which mainly utilizes text as the medium of information.

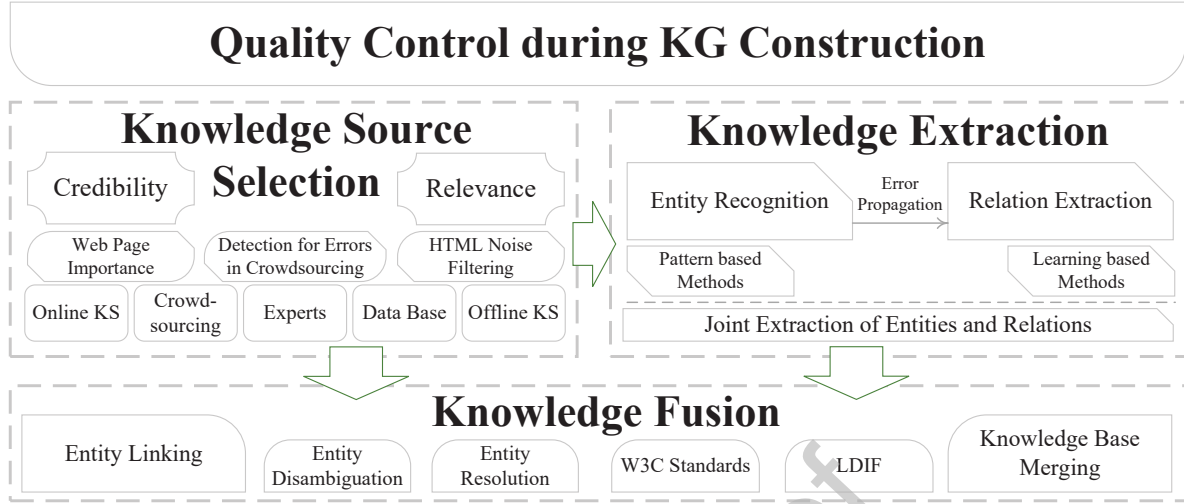


Figure 3: Quality control during KG construction

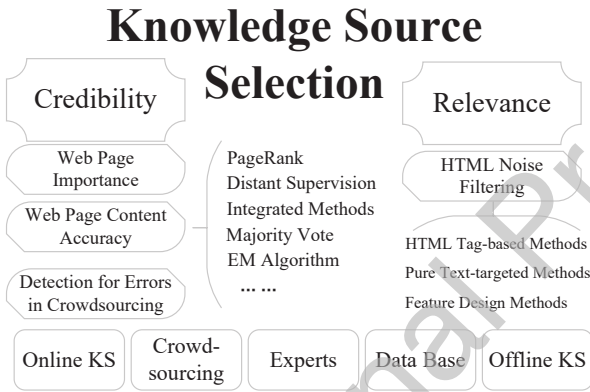


Figure 4: Schema of knowledge source selection

or extracting knowledge from web pages⁶. Web pages contain noise in the form of advertisements and unwanted scripts with low correlation with the knowledge of interest, leading to a reduction in the relevance of the knowledge source. Web pages exhibit different credibility levels. Fogg *et al.* [47] suggested that the level of commercial characters and entertainment may have negative influence on the credibility of websites. For example, government and education websites are generally more credible than commercial or entertainment websites. Therefore, it is important to evaluate the credibility of web page and filter out noise for high-quality knowledge source selection.

(1) Online Knowledge Source Credibility Evaluation. Generally, the credibility of web pages is evaluated from the perspectives of importance and content accuracy

[48]. From these perspectives, the importance of web pages can be evaluated through link-based information in the web structure, and the accuracy of web page content can be evaluated through verification by an external knowledge base.

Web page importance is generally represented through link-based information of web structures. Link-based evaluation methods for web page importance are mainly designed through a propagation mechanism. For example, PageRank [49] is a classic web page importance evaluation algorithm based on a propagation mechanism. It utilizes the “PR value” metric to represent web page importance. Because the importance of web pages reflects the credibility of knowledge sources, this link-based evaluation method can be used as a heuristic for evaluating the credibility of knowledge sources.

There are some improved propagation-based algorithms for the evaluation of web page importance [48, 50, 51] and link data credibility [52]. Generally, these methods construct a network structure with links between different web pages or data and evaluate the credibility of target knowledge sources through propagation of the designed metrics. However, link-based information is not fully available on some web pages, which sometimes makes it difficult to evaluate their importance [53].

The accuracy of web page content is also an important perspective for evaluating the credibility of knowledge sources. The evaluation methods for accuracy are mainly designed using distant supervision [54]. Generally, the content of web pages is compared with an external knowledge base for verification, and a metric is further deployed to evaluate the accuracy of the knowledge source.

In addition, some methods integrate various characteristics of web pages to comprehensively evaluate credibility. For example, Olteanu *et al.* [55] proposed a method to as-

⁶“Web page” refers to any online knowledge source here.

sess web credibility by investigating various characteristics of web pages, such as textual content, link structure, and web pages design. This method selects the most informative features through statistical analyses and infers the credibility of knowledge sources by employing supervised learning algorithms, which provides a comprehensive credibility evaluation framework.

(2) Online Knowledge Source Relevance Enhancing. In addition to credibility, the relevance of online knowledge sources is important for KG quality. As described earlier, the noise in web pages may contain uninteresting or incorrect information. Therefore, filtering out noise in HTML text⁷ is important for enhancing the relevance of web knowledge sources.

Noise filtering methods for HTML text can be mainly divided into 1) HTML tag-based, 2) pure text-targeted, and 3) feature design-based methods. 1) HTML tags formulate the structure of content on web pages. Some noise can be distinguished directly from the HTML tags, such as script code, comments, and advertisements. Another portion of noise can be distinguished using statistical methods combined with HTML tags. For example, Yi *et al.* [56] proposed the style tree to capture HTML content reputation according to HTML tags. According to the authors, noise often shares similar tag styles. Based on this observation, the authors evaluated the relevance of each node in the style tree through entropy-based measures and filtered out the nodes with low relevance. 2) In addition to HTML tag-based methods, there are filtration methods for pure text content. For instance, HTML tags can be removed, and noise filtration can be applied to pure text using word frequency [57]. 3) As an example of the third category, a vision-based HTML structure was constructed, and multi-dimensional features were designed for noise filtration [58]. Debnath *et al.* [59] partitioned a web page into constituent web page blocks and designed specialized functions to distinguish noise in web page content.

3.1.2. Crowdsourcing Knowledge Source Selection

Crowdsourcing is an intensive knowledge acquisition method that generally assigns the same knowledge construction tasks to different workers and obtains different results from them. Currently, many well-known platforms for knowledge-intensive crowdsourcing (KIC)⁸ have been developed. Furthermore, many well-known large-scale KG construction projects, such as Wikidata [60] and Freebase [61], use crowdsourcing for ontology construction⁹.

⁷Web page content may also include pictures, videos, audio, etc. The web page content discussed in this paper mainly refers to information in the form of text. Among such text, HTML text is the most important form.

⁸The current well-known KICs mainly include Mechanical Turk of Amazon and Ali crowdsourcing

⁹A type of KG construction of concepts, which consists of entities and relations that conform to existing concepts.

Crowdsourcing is used for knowledge source construction. The credibility of this type of knowledge source is determined by the selection of knowledge that different workers construct. For example, a labeling corpus is a typical crowdsourcing task wherein the same corpus is assigned to different workers (annotators) for corpus labeling. In fact, the credibility of the labels refers to their accuracy here. Therefore, the selection of high-accuracy labels can be regarded as the selection of high-quality crowdsourced knowledge. One of the most important methods for selecting high-quality crowdsourced knowledge sources is to filter out false labels.

The false labels in crowdsourcing mainly come from “spammers”¹⁰ [62] and mistakes made by multiple workers. These false labels reduce the credibility of crowdsourced knowledge, thus degrading KG quality in terms of, e.g., trustworthiness and accuracy. The effective detection of spammers and mistakes is an important step for quality control in KG construction. Therefore, we briefly introduce some methods as follows:

- (1) The majority vote is a widely used crowdsourcing label aggregation method that favors the most frequent label. This type of method adopts the rule of “the minority obeys the majority”, exhibiting good performance when the accuracies of different annotators are close. Furthermore, it avoids the mistakes of normal workers when there are few “spammers”. However, the majority vote lacks the ability to detect “spammers”. One way to address this issue is to use machine learning (ML)-based methods to filter out spammers and utilize majority vote on the remaining workers [62]. Another issue with the majority vote method is that it does not consider the individual level of annotators and is therefore less effective in situations where there is a gap in label accuracy among different annotators.
- (2) Compared to majority voting, the expectation-maximization (EM) algorithm [63] considers the individual level of annotators through the statistical distribution of the label accuracy of annotators. The EM algorithm works iteratively to update the parameters of the distribution through likelihood estimation and obtains the statistical distribution of label accuracy. Spammers can be distinguished according to the obtained distribution, and the high-confidence labels are selected. Some enhanced versions of the EM algorithm also exist. For instance, the research in [64] studied the lack of separating capacity for “unrecoverable true errors” and “recoverable bias” of annotators in EM algorithms. This method extracts these two features from the confusion matrix of the EM algorithm and enriches the dimensions of the annotator description, leading to an improvement in high-quality label selection.

¹⁰A “spammer” is an extremely bad worker, such as a random annotator or unified annotators.

- (3) Some statistical models are often used together with EM algorithms for parameter calculation. For example, an empirical Bayesian algorithm iteratively eliminates spammers and estimates the consensus labels based on good annotators only [65].
- (4) Labeling the sequence tags is an important labeling task in crowdsourcing. For example, the boundary of entities in a corpus is usually labeled through a sequence of word tags. The hidden Markov model (HMM) is widely used for this type of crowdsourcing labeling task. In the sequence tagging task, the position of the to-be-tagged element influences its type, which in turn requires a sequence. In this case, the sequence feature representation ability of the HMM is effective for the selection of sequence tagging results. Furthermore, the HMM can be combined with some recurrent neural network (RNN) models, such as long short-term memory (LSTM) and models that learn in the model space [66–73], to predict a non-tagged sequence. This leads to the idea of less manual work consuming methods for crowdsourced sequence knowledge construction [74].
- (5) Some neural network (NN)-based methods are also used in crowdsourcing label selection. NN-based methods use an NN to automatically learn the features of the annotators and labels and further select high-confidence labels. For instance, bidirectional LSTM (Bi-LSTM) learns annotator-generic and -specific features for label accuracy evaluation [75]. This idea is similar to the “true error & worker bias” used in the improved EM algorithm [64] discussed above, but it uses different models.

It is evident that there are a variety of models targeting feature analyses for high-accuracy crowdsourcing label selection.

3.1.3. Offline Knowledge Source Selection

Not all knowledge sources originate from accessible web pages or crowdsourcing construction. Some knowledge sources may be offline¹¹, such as books and medical records. For offline knowledge sources, selection focuses on text content. In this case, the quality control of offline knowledge sources is similar to that of web page text, including relevance control between the target text, knowledge of interest, and credibility evaluation of the text content. Quality control methods for web content are also used to improve the quality of offline knowledge source-based KG construction.

3.1.4. Database Knowledge Source Selection

A database is an important source of data for the construction of KGs. Compared with other data sources, most

¹¹An offline knowledge source may be cross-media, including knowledge with the storage of sound, text, or vision. Because cross-media sources are not the focus of this paper, here, offline knowledge refers to that in a text format only.

Knowledge Extraction

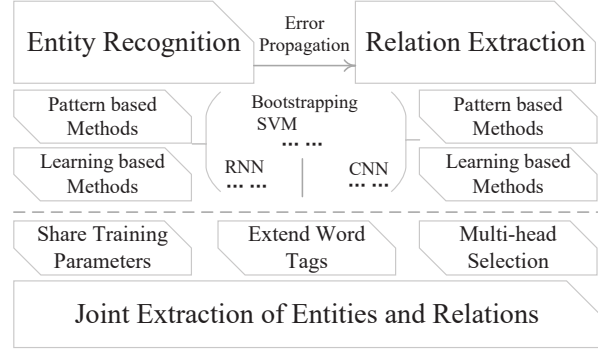


Figure 5: Schema of knowledge extraction

data in a database are presented in a structured or semi-structured format. For some offline databases, distant supervision methods, which use some public large-scale ontology databases or domain knowledge bases for quality control, can be used for evaluation [61].

3.1.5. Experts as a Knowledge Source

Some domain KGs are directly designed by experts, and their “knowledge source” is expert knowledge, which cannot be evaluated directly. In some cases, distant supervision methods can be used to evaluate the quality of construction results [76], and manual inspection methods can be used for quality control.

3.2. Knowledge Extraction

The knowledge extraction process extracts high-quality knowledge elements from knowledge sources. Knowledge extraction determines the degree to which a KG follows knowledge sources; therefore, quality control of knowledge extraction is an essential step in KG quality control during construction.

Generally, knowledge extraction extracts structured knowledge from semi-structured or unstructured data [132]. It consists of two main processes: named entity recognition (NER) and entity relation extraction (ERE) [133]. NER and ERE are prone to quality problems, such as semantic drift, errors in entity extraction, and errors in relation extraction. These problems could negatively affect the accuracy, trustworthiness, and completeness of a KG. The quality control of knowledge extraction must solve these problems while ensuring the quality of the extracted knowledge. Quality control of knowledge extraction is important to maintain KG quality. Figure 5 illustrates the schema of knowledge extraction.

3.2.1. Quality Control of NER

NER extracts concept mentions of interest from the knowledge source. The results of NER might affect the

Table 6: References to knowledge source (KS) selection

Online KS Selection	Credibility Evaluation	Page <i>et al.</i> [49] Kleinberg <i>et al.</i> [50] Langville <i>et al.</i> [48]	Nguyen <i>et al.</i> [52] Popat <i>et al.</i> [54] Olteanu <i>et al.</i> [55]
	Relevance Enhancing	Yi <i>et al.</i> [56] Gibson <i>et al.</i> [57]	Liu <i>et al.</i> [58] Debnath <i>et al.</i> [59]
Crowdsourcing KS Selection		Lease <i>et al.</i> [62] Ipeirotis <i>et al.</i> [64] Raykar <i>et al.</i> [65]	Nguyen <i>et al.</i> [74] Yang <i>et al.</i> [75] Ipeirotis <i>et al.</i> [64]
Offline KS Selection		Popat <i>et al.</i> [54]	
Database KS Selection		Nguyen <i>et al.</i> [52]	
Experts as KS			

quality of the KG construction steps, such as ERE and knowledge fusion. In the process of NER, the error-recognized entities lead to inaccurate knowledge in KGs, thereby degrading KG accuracy and trustworthiness. The unrecognized entities negatively affect the KG completeness. Therefore, the quality control of NER is crucial during KG construction. From the perspective of the impact on KG quality, NER methods can be divided into pattern-based and learning-based methods.

Quality Control of Pattern-based NER Methods. Early pattern-based NER methods are mainly based on rule design. These methods generally focus on the pattern features of the letter structure of entities. The effectiveness of such methods depends on the quality of the rule design. In other words, the quality of the rule design of entities determines the quality of entities, thus influencing the KG quality. For effective rule design, domain adaptation of rule-based annotators for NER has been explored to formulate an efficient framework for manual rule design [77].

In addition to rule design methods, the iterative extraction of entities from a knowledge source is also a popular choice [134]. For example, bootstrapping is a representative algorithm of this type [79]. By selecting some entities and patterns as seeds, it automatically extracts new entities and discovers new patterns from the knowledge sources, which are merged into the seeds for the next iteration of new instance extraction.

Iterative entity extraction methods automatically discover entities from large-scale knowledge sources. However, semantic drift during the iterative process poses a challenge [18]. For example, if the knowledge of interest is “animal”, a selected seed “chicken” can refer to “animal” and “meat” at the same time. When it appears as “meat”, some irrelevant entities may be extracted due to their co-occurrence with “chicken”, such as “beef” and “vegetables”. This situation leads to degradation of KG accuracy. To address this problem, some techniques have been designed, including type checking, instance confidence assess-

ment, and graph-based algorithms. We briefly introduce these methods as follows:

- (1) The type checking method compares the type of extracted instance with the category of patterns of interest and filters out the conflicting ones [82, 84]. For example, if the type of extracted instance “Einstein” is “Person”, which does not conflict with the pattern category “Physicist”, then “Einstein” is accepted. However, if the pattern category is “Discipline”, then “Person” and “Discipline” are mutually exclusive, and “Einstein” is filtered out. However, type checking cannot distinguish fine-grained semantic errors. For example, for the pattern category “Doctor”, the extracted instance “Einstein” is of type “Person”. Because “Doctor” and “Person” do not conflict, this error cannot be filtered out.
- (2) The instance confidence assessment method is similar to the quality assessment of knowledge sources. This method evaluates the confidence of the extracted instances or new patterns and filters out the low-confidence instances and patterns [78, 80]. Confidence evaluation generally utilizes heuristic methods, which consider factors such as the frequency of extracted instances, occurrence of patterns in the seed set, and credibility of knowledge sources.
- (3) The graph-based algorithm on semantic drift constructs a graph with connections between patterns and instances, and it evaluates the confidence of instances or patterns through graph-based algorithms, such as random walk [83] and regularized Laplacian. Finally, it selects high-confidence patterns and instances [81].
- (4) The above methods may reduce the recall of entities while filtering noise out, thereby compromising the KG completeness. To combat this issue, “drift points” can be identified to remove information extraction errors, while multi-concept instances remain intact as noise is filtered out [18].
- (5) In recent research, a crowd-powered approach was proposed to diminish the effects of semantic drift [135].

Table 7: References to knowledge extraction

Named Entity Recognition	Pattern-based Methods	Chiticariu <i>et al.</i> [77] Thelen <i>et al.</i> [79] Li <i>et al.</i> [18] Pasca <i>et al.</i> [82] Curran <i>et al.</i> [84] Stavropoulos <i>et al.</i> [86] Meroño <i>et al.</i> [88]	Riloff <i>et al.</i> [78] Pantel <i>et al.</i> [80] Komachi <i>et al.</i> [81] Fouss <i>et al.</i> [83] Wang <i>et al.</i> [85] Stavropoulos <i>et al.</i> [87]
	Learning-based Methods	Zhou <i>et al.</i> [89] Malouf <i>et al.</i> [91] Li <i>et al.</i> [92] Rocktaschel <i>et al.</i> [94] Liu <i>et al.</i> [96] Bengio <i>et al.</i> [98] Huang <i>et al.</i> [100] Zhou <i>et al.</i> [102] Rodriguez <i>et al.</i> [104] Yang <i>et al.</i> [106] Shang <i>et al.</i> [108] Cao <i>et al.</i> [110]	Ma <i>et al.</i> [90] Yadav <i>et al.</i> [19] Li <i>et al.</i> [93] Fang <i>et al.</i> [95] Lin <i>et al.</i> [97] Ratner <i>et al.</i> [99] Jia <i>et al.</i> [101] Jia <i>et al.</i> [103] Cao <i>et al.</i> [105] Yang <i>et al.</i> [107] Ni <i>et al.</i> [109] Liang <i>et al.</i> [111]
Entity Relation Extraction	Pattern-based Methods	Brin <i>et al.</i> [112] Agichtein <i>et al.</i> [113]	
	Learning-based Methods	Kambhatla <i>et al.</i> [114] Zhou <i>et al.</i> [116] Bunescu <i>et al.</i> [117] Collins <i>et al.</i> [119] Zeng <i>et al.</i> [121] Hong <i>et al.</i> [123] Geng <i>et al.</i> [125] Fei <i>et al.</i> [127]	Zeng <i>et al.</i> [115] Lin <i>et al.</i> [97] Hasegawa <i>et al.</i> [118] Lin <i>et al.</i> [120] Chen <i>et al.</i> [122] Ji <i>et al.</i> [124] Li <i>et al.</i> [126]
Joint Extraction of Entities and Relations	NN-based Methods	Miwa <i>et al.</i> [20] Zheng <i>et al.</i> [129] Zeng <i>et al.</i> [131]	Bekoulis <i>et al.</i> [128] Zeng <i>et al.</i> [130]

The coupled pattern learner (CPL) was proposed, and the learning capabilities of the CPL were improved by asking people to take part in the knowledge acquisition process. Concept drift, which focuses on the drift of concepts over time¹², is also widely researched [85]. Identity- and morphing-based approaches have also been introduced for the drift of concepts over time, becoming the main methods in this field. On this basis, some improved methods and visual tools have been proposed to serve entity recognition and improve the quality of extracted entities and concepts [86–88].

Quality Control of Learning-based NER Methods. Learning-based NER methods learn the features of word sequences and make predictions on word tags for the

boundary recognition of entities [134]. The quality of NER in these methods depends on the 1) feature and model design and 2) quality of the tagged corpus.

In learning-based NER models, features are designed to represent entity characteristics. Learning-based models make predictions on entity boundaries by learning features. From the perspective of feature learning approaches, learning-based models can be divided into traditional ML systems and NN learning systems.

The NER quality of a traditional ML system is mainly determined by the feature design. Traditional ML systems generally learn the features of each word from the designed feature space. Thus, the selected dimensions of the feature space are essential for this type of model. Zhou *et al.* [89] designed a feature space consisting of 11 orthographic features, entity trigger words, and gazetteer words and utilized the HMM for the classification of tags. Malouf *et al.* [91] conducted a more refined design of the feature space by adding features such as capitalization, first occur-

¹²For example, the concept of European Union was developed from European Community with the passage of time. In this case, concept drift occurred from European Community to European Union.

rence flag of words, occurrence flag with a known last name and first names from dictionaries. This method uses the maximum entropy (ME) model for tag classification and achieved better performance through a more refined design of features. In addition to the feature design perspectives mentioned above, there are word feature dimensions, such as neighborhood words [92], suffix features [94], and word embedding features [96].

The NER quality of an NN learning system is determined by both the feature and model design. An NN learning system learns hidden features from the word sequence and some extra sequence attributes directly. The recurrent NN (RNN) is widely used in NN learning systems of NER. Many methods improve the effectiveness through better model design. For example, to better learn the semantic features contained in a character sequence, the authors of [98] proposed an RNN-based LSTM model that considers the long- and short-term features of word sequences. Another study [100] utilized Bi-LSTM and added a conditional random field (CRF) to the top layer, which further improved the NER performance. In addition, there are some RNN methods that consider more refined semantic features, such as character-level features [90] and affixed features [19], as auxiliary features of NER, which improves the model effectiveness through better feature design.

Recently, cross-domain NER has received increased research attention due to the high cost of manual labeling [101]. Based on NN learning, transfer learning has achieved great success in cross-domain NER tasks, including multi-task learning [102, 103], fine-tuning [104], share-private [105], and knowledge distilling [106]. Transfer learning methods recognize and apply knowledge and skills learned in previous tasks to novel tasks through the training of labeled data in a new domain on an NN that was trained for the original domain. In this manner, the labeling cost is greatly reduced, and the NN model has a stronger ability to recognize entities in various domains. Furthermore, knowledge of higher quality and larger scale can be obtained.

In addition to the model and feature design, high-quality tagged corpus construction is also crucial to the effectiveness of learning-based NER methods. A large-scale manually tagged corpus is extremely laborious. Therefore, some automated corpus tagging methods have been proposed to build high-quality tagged corpora with low labor costs.

In recent years, automated corpus tagging methods have mainly been based on distant supervised methods together with weakly supervised methods:

- (1) Distant supervised methods for corpus tagging generally compare structured data of an external knowledge base with unstructured texts of a corpus and perform tagging according to information of the external knowledge base, such as entity boundaries and entity types. Although this method is low-cost, it leads to incomplete or erroneous labeling due to be-

ing limited by the quality of the external knowledge base. More accurate tagging can be achieved through distant supervision; however, it is still unable to completely eliminate tagging errors [93, 95]. To fill this gap in distant supervision, an attention mechanism was introduced to recognize noise in the training data [97]. Recent studies have attempted to deal with label incompleteness and noise issues by enhancing the quality of distantly supervised data with various methods. Some researchers have adopted partially annotated CRFs to consider possible labels for incomplete labels [107, 108]. Some methods are designed to filter out noise, such as heuristic rules on matching quality [109] and occurrence popularity-based methods [110], which are called weakly supervised methods and are introduced in more detail below. In addition, a specialized learning model was designed to address the above issues in distantly supervised labeled corpora [111].

- (2) Weakly supervised methods for corpus tagging generally utilize multiple, possibly conflicting, rules to label training data, resulting in large-scale label instances with unknown accuracy. Some heuristic algorithms have been further applied for high-quality tag selection. With this type of method, it is possible to obtain large-scale weakly supervised tagging data with less manual work and integrate them for high-quality tag generation. Snorkle [99] is a commonly used weakly supervised framework for data labeling. It obtains weakly supervised labeled data through user-designed low-quality rules. Complementary weakly supervised data are subsequently chosen, and high-quality labeled data are generated.

3.2.2. Quality Control of ERE

ERE utilizes the recognized entities in NER and extracts the relations between these entities through semantic analyses, thereby constructing the semantic structures of KGs. The results of ERE directly affect the accuracy and completeness of the KG. The errors in the extracted relations degrade the accuracy of the KG and further reduce the KG trustworthiness, and the missing extraction of existing relations harms the KG completeness. Therefore, the quality control of ERE is essential for KG quality. Similar to NER, ERE methods can also be divided into pattern-based and learning-based methods according to their influence on KG quality.

Quality Control of Pattern-based ERE Methods. Early pattern-based methods of ERE extract specialized relationship patterns through expert rules, requiring significant manual work for rule design. Semi-supervised ERE methods enable pattern-based bootstrapping, which helps reduce the cost of rule-based methods. Bootstrapping algorithms for ERE iteratively discover new patterns and new entity pairs that match the relation, starting with

a few instances of the relation of interest as seeds. One of the earliest algorithms is dual iterative pattern relation expansion (DIPRE) proposed by Brin [112]. As mentioned for NER, the evaluation of extracted instances is important for the accuracy of bootstrapping-based methods. However, DIPRE ignores the possible errors that occur during bootstrapping and lacks an instance evaluation process. To combat this issue, Snowball [113] focuses on the quality control of bootstrapping ERE methods. Snowball designs more refined patterns, including named entity tags (PER, ORG, LOC, etc.), in the patterns to reduce errors during iterated extraction. For instance evaluation, Snowball deploys a method to evaluate the confidence of tuples and patterns to discard instances with low confidence values.

Quality Control of Learning-based ERE Methods. Learning-based relation extraction models generally focus on the semantic features of sentences. Similar to learning-based NER methods, the feature design, model design, and quality of labeled data determine the quality of learning-based ERE methods. Learning-based ERE methods also include 1) traditional ML systems, 2) NN systems that are supervised and share similar characteristics to those of NER, and 3) unsupervised ERE methods.

The traditional ML systems of ERE design the features of relations and utilize a classifier for relation classification. 1) Generally speaking, a more refined design of the feature space leads to better classification performance. For example, Kambhatla [114] trained an ME model with 49 relation classes based on some lexical and syntactic features. Zhou *et al.* [116] designed more dimensions of features and improved the relation classification performance. 2) Furthermore, to avoid complex feature engineering, kernel methods are also used in ERE. The features learned in kernel methods are represented by the selected kernel functions, which directly calculate the similarity between two relations. For example, the syntactic tree kernel method computes the common subtrees shared by two syntactic parse trees to calculate the similarity between two relations. Then, the relations are classified according to the similarities. In kernel methods, the kernel function selection is the feature design process, which includes the sequence kernel [117] and syntactic tree kernel [119].

The NN systems of ERE are designed to learn semantic relation features hidden in a word sequence. While the NN systems of NER are concerned with the entity features and must predict the tags of each word, the NN systems of ERE focus on the semantic relation between entities, which is a global feature of a word sequence. ERE must only predict the classification of the relation of each word sequence. In response to these characteristics, convolutional NNs (CNNs) are widely used in ERE tasks and have good global feature learning ability. Some studies have focused on the improvement of a CNN in ERE tasks. Zeng *et al.* [121] improved the ability of CNNs to learn semantic features through word embeddings and location informa-

tion integration. Zeng *et al.* [115] extended this work further and proposed the piecewise CNN (PCNN) model. This study extended the accurate pooling area from one sentence to a segment of sentences, improving the effectiveness of ERE. For quality control of labeled data, Lin *et al.* [97] introduced the attention mechanism to the ERE model so that the model can identify important sentences while filtering out noise, which enhances the quality of the training data.

Recent studies on ERE have proposed some novel NN structures with an attention mechanism to reduce the effect of noise in the lexical sequence, usually utilizing distant supervision to automatically find new relations [122–124]. Geng *et al.* [125] proposed a sequential and tree-structured LSTM with attention based on a dependency tree. By considering both structural and word-based features, they improved the quality of relation extraction. Li *et al.* [126] proposed a novel selective gate framework to deal with noise in distantly supervised training data. This framework develops a self-attention mechanism to capture rich contextual dependencies, and based on this, it designs a pooling-equipped gate instead of selective attention. In addition, the extraction of overlapping entity relations has received extensive research attention recently, considering that multiple relations may occur in one sentence between various pairs of entities. This issue is solved with end-to-end models [127], as discussed in “Joint Extraction of Relations and Entities” below.

The aforementioned supervised ERE systems depend on labeled data, so low-quality labeled data negatively affect their performance. To address this issue, some unsupervised ERE methods based on clustering have been proposed. One of the earliest approaches to completely unsupervised ERE was proposed by Hasegawa *et al.* [118]. Two named entities are said to co-occur if the number of intermediate words between them is less than a threshold value. The co-occurring entities and intermediate words between them are recorded and transformed into word vectors to be classified using clustering methods. Furthermore, some clustering methods exploit dependency paths. Lin *et al.* [120] argued that expressions with similar dependent paths usually express the same relation, so the authors proposed the discovery of inference rules (DIRT) algorithm, which finds named entity pairs with the same relation by discovering similar dependency paths.

Joint Extraction of Relations and Entities. The ERE process generally utilizes the outputs of entities of NER processes [136] in a serial manner. The errors generated in NER are propagated to ERE, leading to error accumulation and further degradation of KG quality. To avoid the accumulation of errors, joint extraction methods for entities and relations have been investigated. The idea of joint extraction is to perform the two processes with a fusion model, thereby reducing the accumulation of errors. Recent joint extraction methods can be roughly di-

vided into table filling, tagging, and sequence-to-sequence (Seq2Seq) methods [130]. Some NN-based joint extraction methods are introduced as follows:

- (1) Sharing the training parameters of the NNs of NER and ERE is an effective method for reducing error accumulation; this can be utilized in table filling methods, which require the model to enumerate the possible entity pairs. Miwa *et al.* [20] designed an end-to-end joint extraction NN architecture. They designed four embedding layers to share semantic information in the NER and ERE tasks. The NER component uses a Bi-LSTM layer with a fully connected layer for sequence labeling. The ERE component uses Bi-LSTM to capture entity relation features in the dependency tree. The prediction tags of the NER are encoded through one of the embedding layers and passed to the ERE layer. The relevance between the two tasks is strengthened by sharing the training parameters and fusing the NER prediction with ERE, thereby reducing the error accumulation of serial processes.
- (2) Extending the information of word tags, which is a type of tagging method, can enrich the semantic information of word sequences. Extended word sequence tags can express the relation features together with entity features, leading to the fusion of two serial processes into one common task. Zheng *et al.* [129] transformed the joint extraction task into a sequence tag prediction task by extending the semantic information of word tags. The word sequence tags in NER only contained word features, such as parts of speech and type. The authors added information on relation categories and the role of entities to the word tags. In this manner, the relation features were incorporated into the sequence information, greatly improving the performance and reducing the error accumulation. However, it cannot deal with overlapping relations because the model with tagging methods cannot assign different relation tags to one token.
- (3) Relationship overlap¹³ in sentences may result in the misidentification of relations in the extraction process [127]. To address this issue, Bekoulis *et al.* [128] aimed at the possible relation overlap problem and treated it as a multi-head selection task. Based on the Bi-LSTM-CRF structure, the relations between entities of a sentence were predicted to avoid relation overlap, and joint extraction led to improved KG completeness.
- (4) Recent studies on joint extraction of entities and relations have also introduced Seq2Seq models, which receive unstructured sentences as input and decode entity relation triples directly as a sequential output [131]. Zeng *et al.* [130] developed a Seq2Seq-based joint extraction method called CopyRE, which expands a Seq2Seq framework with a copy mechanism in the

decoder. The authors focused on two main drawbacks of CopyRE: instability and inability to extract entities with multiple tokens. The improved method, CopyMTL, allowed the model to predict multi-token entities.

3.3. Knowledge Fusion

Knowledge fusion is the process of revising errors in KGs and integrating KGs and external knowledge bases [137]. Knowledge fusion is generally divided into two categories: entity linking and external knowledge base merging. In many cases, the knowledge fusion process is included to deal with some KG quality issues that are not focused on during the selection and extraction processes.

3.3.1. Entity Linking

Entity linking (EL) refers to linking an entity to a corresponding entity in an external knowledge base¹⁴ through entity mentions and the corresponding context information. This process can filter out redundant and incorrect entities and confirm the correct concept of ambiguous entities to improve KG quality; the main parts related to knowledge quality control include entity disambiguation and co-referential resolution.

Entity disambiguation is a way to resolve the ambiguity of entities with the same entity mention. For example, the mention “Apple” may represent the fruit apple or the company Apple. It is necessary to determine which type of entity the mention refers to for accurate semantic expressions in KGs. Generally, the methods for entity disambiguation are based on similarity analysis, which designs some types of metric for the similarity of the entity mention “Apple” and entity objects, i.e., the fruit apple or the company Apple. Then, the entity mention is linked to the entity object with a higher similarity metric value. Traditional similarity metric design methods can be mainly divided into four groups: 1) Bag-of-words models [138] utilize neighborhood words as vector representations and calculate the cosine similarity. 2) Semantic models [139] integrate semantic features into bag-of-words models, such as semantic features of a text space. 3) Social network models [140] construct a structure of mentions through relations and calculate topological distance between mentions as the similarity. 4) Encyclopedia knowledge models calculate similarities through the link information of encyclopedias.

Entity resolution is the process of solving the problem of multiple mentions referring to the same entity object. For example, “Barack Obama” and “President Obama” may refer to the same entity object, and some pronouns “he” and “him” in the text may also refer to it. These mentions must be associated with the corresponding entity

¹³ “Relationship overlap” refers to the possibility that the same entity may have multiple relationships with other entities in a sentence.

¹⁴ “Knowledge base” refers to a high-quality external knowledge base, and it is generally believed that the knowledge base correctly contains the entities of interest.

object to eliminate redundancy and improve the KG quality. Early technologies for this task are based on syntactic analysis, such as the Hobbs algorithm [141] and centripetal theory [142]. With the introduction of ML, methods such as decision tree and the Dempster-Shafer model [143] are used in co-referential resolution. In addition, clustering-based methods match mentions and entity objects through similarity calculation methods, such as pointwise mutual information [144].

With the great success of NNs in NLP, recent studies on EL have tended to utilize NNs to automatically learn the similarity between surface mentions and entities. Traditional EL methods manually design entity features for similarity evaluation, which requires professional domain knowledge and massive labor cost. In contrast, deep NNs (DNNs) can learn the representation of entity mentions in a contextual document, which are also referred to as entity embeddings [145]. Subsequently, the embeddings of entities and entity mentions are obtained to calculate the similarity for entity linking. Later, end-to-end models were proposed to jointly process NER and EL to reduce pipeline errors [146]. Ganea *et al.* [147] pioneered the reliance on entity embeddings, the attention mechanism for learnable context representation and differentiable joint inference, to learn basic features as well as their combinations from scratch. Their method jointly extracts entity mentions and links them to suitable entities. Furthermore, a series of follow-up studies have presented potential solutions to improve the effectiveness of EL [148–151].

3.3.2. Knowledge Base Merging

From the perspective of knowledge fusion, entity linking could be regarded as the merging of entities between a high-quality knowledge base and a KG in construction, wherein knowledge base merging includes the linkage of both entities and relations in a KG. Using high-quality knowledge bases to merge knowledge is helpful to supplement the knowledge that is missing in the knowledge source, thereby enhancing KG completeness. Merging high-quality knowledge is also conducive to improving the KG accuracy and trustworthiness.

Recently, the field of multi-source knowledge fusion has received increasing research attention, employing NN-based methods to support the knowledge base merging [152]. Knowledge base merging requires the correct linking of entities and relations between different knowledge bases. The general goal is to calculate the similarity between various entities (relations) and choose the most similar one for linkage, so a reasonable representation of entities and relations is important. Such methods, wherein different deep NN-based encoders are used, are employed for entity, entity attribute, and relation embedding [153–155]. A similarity calculation method based on semantic labels was also proposed for a more explainable model with designed features [156].

In addition, to facilitate the mutual conversion between

relational database and RDF format¹⁵, W3C proposed two conversion standards: direct mapping and R2RML. These frameworks and standards can assist in the merging of knowledge to improve KG quality.

4. Enhancement of KG Quality

Although the quality control deployed in each process of KG construction reduces the negative influence on KG quality to an extent, there still exist some quality problems once the KG has been constructed, including the following: 1) Some errors inevitably occur during KG construction, such as missing entity or relation and wrong types of entities or relations. 2) The timeliness of some KGs gradually decreases over time, especially for KGs regarding quickly changing domains, such as user interest prediction and dynamic social networks. 3) New requirements for KG quality may arise in their application, leading to the need for KG expansion. Therefore, it is necessary to correct, update, and complete KGs to enhance their quality. This section discusses KG quality enhancement methods from the above perspectives.

4.1. Accuracy Enhancement of KG

KG accuracy reflects the degree to which knowledge follows facts and is the most important dimension of KG quality. The main factors degrading KG accuracy are wrong relations, wrong entities, and wrong attributes in the KG. In response to this, error detection should be performed for accuracy enhancement.

Error detection is the task of finding and correcting knowledge a KG that is inconsistent with objective facts in. According to negative factors for KG quality, error detection methods are divided into error detection of relations, error detection of entity type, and error detection of attributes [2].

4.1.1. Error Detection of Relations

The goal of error detection of relation is to find relation assertions in the KG that do not match facts, including detection with internal and external knowledge.

Internal knowledge-based detection methods utilize the knowledge from the KG itself and judge the relations by exploring the association features or distribution of the entities in the graph. For example, Lao *et al.* [157] proposed the path ranking algorithm (PRA), a typical detection method that uses the paths between entity pairs to discover associated features. The PRA uses multiple random walks to construct features and infers the relation between two entities to achieve error detection of relations.

In addition to the path information of a KG, the types of entities are used for error detection of relations. Melo *et al.* [23] considered the subject types and object types

¹⁵RDF is the resource description framework used by the LOD knowledge base.

Table 8: References to the enhancement of KG quality

Accuracy Enhancement of KG	Relation Assertions Error Detection	Lao <i>et al.</i> [157] Paulheim <i>et al.</i> [158] Paulheim <i>et al.</i> [160]	Melo <i>et al.</i> [23] Dong <i>et al.</i> [159]	
	Type Assertions Error Detection	Paulheim <i>et al.</i> [158]		
	Literal Value Error Detection	Wienand <i>et al.</i> [161] Golab <i>et al.</i> [163] Fan <i>et al.</i> [165]	Fleischhacker <i>et al.</i> [162] Koudas <i>et al.</i> [164] Li <i>et al.</i> [35]	
Timeliness Enhancement of KG	Global Update of KGs	Auer <i>et al.</i> [166]		
	Local Update of KGs	Hellmann <i>et al.</i> [167] Pinsky <i>et al.</i> [169] Liang <i>et al.</i> [22]	Winkler <i>et al.</i> [168] Cho <i>et al.</i> [170] Jiang <i>et al.</i> [171]	
Completeness Enhancement of KG	Completion of Relations	Balavzevic <i>et al.</i> [172] Chen <i>et al.</i> [174] Wang <i>et al.</i> [1] Ji <i>et al.</i> [176] Trouillon <i>et al.</i> [178]	Richardson <i>et al.</i> [173] Bordes <i>et al.</i> [175] Lin <i>et al.</i> [21] Nickel <i>et al.</i> [177] Shi <i>et al.</i> [179]	
		Completion of Entity Types	Paulheim <i>et al.</i> [180]	
		Completion of Attributes	Razniewski <i>et al.</i> [181] Liang <i>et al.</i> [182]	
Consistency Enhancement of KG		Gangemi <i>et al.</i> [183] Paulheim <i>et al.</i> [160] Tpper <i>et al.</i> [186]		Ma <i>et al.</i> [184] Jang <i>et al.</i> [185]
		Availability Enhancement of KG	Bonifati <i>et al.</i> [187] Lissandrini <i>et al.</i> [189] Hu <i>et al.</i> [191] Lissandrini <i>et al.</i> [193]	
Trustworthiness Enhancement of KG	Zaveri <i>et al.</i> [14] Livraga <i>et al.</i> [194] Chen <i>et al.</i> [151]		Zhao <i>et al.</i> [152] Nguyen <i>et al.</i> [52] Shao <i>et al.</i> [195]	

of relations. The authors combined the type feature and path feature into a relation classifier to detect incorrect relations in a KG.

Statistical distribution-based methods are another representative method that use internal knowledge. Paulheim *et al.* [158] proposed an algorithm that uses the statistical distribution of entity attributes and types to identify wrong relations and improve knowledge quality. The algorithm calculates the distribution of the head and tail entities of a relation in a triplet to infer the probability that the triplet is correct.

Distant supervision is widely used in error relation detection based on external data. Dong *et al.* [159] used the path information of other KGs as prior knowledge to obtain entity-related information for error detection. Paulheim *et al.* [160] mapped the relation and entity types of a triple instance to the corresponding types of an external ontology library. The authors inferred the correctness

of the triples by detecting the relationship of the mapped types. This method is relatively simple to apply and has a wide range of applications.

4.1.2. Error Detection of Entity Type

The error detection of entity type could be regarded as an expansion of error detection of relations, aiming to detect erroneous entity type assertions. The main difference between the two tasks lies in the assumptions about which part is considered correct. More specifically, the task of finding erroneous type assertions assumes that the relations in knowledge triples have a higher accuracy rate than entity type, while error detection of relations often assumes that the entity is correct. Paulheim *et al.* [158] noted that relations have more tendency to be wrong in KGs compared to entity type. Thus, detecting the wrong triple first using a method of detecting wrong relations may be a feasible method for error detection of entity type,

and the corresponding part can be corrected according to different assumptions after the target triple is detected. Therefore, most methods used in error detection of relations can be used in the task of finding erroneous type assertions, including methods based on associated features and statistical distribution.

4.1.3. Error Detection of Attributes

The error detection of attributes is used to find erroneous attributes of an entity. Outlier detection is mainly used in this process of KG accuracy enhancement, aiming to identify the instances in the numerical data that are very different from the majority. The key point in outlier detection is to protect natural outliers, as many correct attributes in a KG tend to be detected as outliers, such as the population of a country or the height of a mountain peak. Wienand *et al.* [161] studied the application of unsupervised numerical outlier detection methods in KGs. The authors used the interquartile range (IQR) and semantic grouping to reduce the detection rate of natural outliers. Fleischhacker *et al.* [162] proposed an extension of this method by combining the outcomes of multiple independent outlier detections.

Erroneous attributes can also be found by detecting multi-attribute consistency. This method detects erroneous attributes by combining the information of entities and discovering inconsistencies. For example, Golab *et al.* [163] used sequential dependencies to express relationships between ordered attributes and find the wrong attributes. Koudas *et al.* [164] studied metric functional dependencies, which allow small differences and can be used for comparison of different attribute formats. Besides sequential and metric functional dependencies, arithmetic operations between numerical attributes can be used for the error detection of attributes. Fan *et al.* [165] proposed a numerical function dependency relationship for detecting wrong numerical attributes, which allows users to specify arithmetic relationships between numerical attributes as data quality rules and detects incorrect attributes in a unified logic framework. Li *et al.* [35] also considered the limitation of the relationship between multiple attributes of entities and proposed a probabilistic framework for detecting errors in the numerical attributes of an entity.

4.2. Timeliness Enhancement of KG

The timeliness enhancement of KGs is mainly achieved through global and local update methods. Global updates maximize the timeliness of KGs. However, as KG scale increases, this method becomes extremely resource-consuming. Compared with global updates, local updates save computing resource consumption but must obtain the required entity information to update. Recently, the local update method has received more attention. We introduce global and local update methods in the following, focusing more on the latter.

4.2.1. Global Update of KGs

Global update methods generally set a time span and perform comprehensive knowledge acquisition after the time span passes, such as knowledge extraction and knowledge fusion. The early updates of some KGs use the global method, such as DBpedia [166]. However, this update method requires frequent network communication and update time because it is almost equivalent to rebuilding the KG. Furthermore, the time span is generally not sufficiently short considering the cost of computing resources, and as a result, some entities may not be updated in time. Therefore, a global update is more suitable for small-scale or low-frequency update KGs.

4.2.2. Local Update of KGs

The key consideration in the local update method is the way the information required for knowledge updates is obtained.

1) One traditional method uses the communication between the knowledge source and KG for local updates. For example, a KG can be passively updated under the direction of the update stream generated during knowledge source changes. This method is adopted by DBpedia, allowing it to update frequently [167]. However, communication with the knowledge source is difficult to implement widely because few knowledge sources support this mechanism.

2) The other traditional local method is to predict the knowledge update, which mainly estimates the update frequency of an entity according to its historical update information. Statistical methods, such as Bayesian inference [168] and stochastic modeling [169], can be used to predict knowledge updates. These methods assume that the update frequency of knowledge satisfies a specific distribution and estimate the update frequency of knowledge based on limited samples. However, Cho *et al.* [170] noted that the complete update history may be unknown in these cases, which contradicts the assumption of the above statistical methods. To combat this, the authors estimated the update frequency of entities in the knowledge source based on the incomplete update history, thereby inferring the synchronization frequency of the KG with the knowledge source.

3) In recent years, the use of deep learning methods for update prediction has gradually become a hot topic. Liang *et al.* [22] noted the inconsistency between the frequency distribution hypothesis of previous work and the actual update frequency of some entities. To solve this problem, the authors designed a predictive model that uses historical update information and semantic information of an entity as features. Jiang *et al.* [171] proposed two time-aware knowledge update prediction models. One model uses temporal order information and imposes time sequence constraints on the geometric structure of the embedded space, while the other time-aware model is based on integer linear programming and considers global time constraints in reasoning. The authors integrated these two

models to further improve the accuracy of knowledge update prediction.

4.3. Completeness Enhancement of KG

KG completeness includes the completeness of relations, entities, and entity attributes. The completeness enhancement of a KG can accordingly be divided into three parts: 1) Completeness enhancement of relations aims to find the implicit relation between entities, which could be regarded as a task to predict missing relations. 2) For entity completion, current methods mainly focus on entity type completion because missing entities are difficult to find without external knowledge. 3) The completion of attributes aims to find the missing attribute values of entities. The knowledge base merging introduced in Section 3.2.2 could be regarded as a method of KG completeness enhancement that uses external knowledge sources. Besides this, there exist internal methods that focus on the internal semantic structure and information of KGs. This section introduces internal methods for each type of completeness enhancement.

4.3.1. Completion of Relations

Completion of relations in KGs aims to predict missing relations between entities. Methods for missing relation prediction mainly include probabilistic, deep learning, and tensor factorization methods [171].

1) Probabilistic models are generally used for completion of relations in the early search. For instance, the probabilistic graphical model [173, 174] sets the correct probability for each knowledge triple to measure the rationality of knowledge. Furthermore, the probabilistic graphical model predicts missing relations using the correct probabilities of knowledge triples in a KG.

2) Deep learning methods learn the latent factor representation of entities and relations for relation prediction. These methods target the proximity of the embedding of the tail entity to that of the head entity, which is transformed by relation to determine whether the triplet is true. TransE [175] is a typical deep learning model used for relation prediction; it treats relations as translations operating in a vector space. It embeds entities and relations into a low-dimensional vector space and uses a custom scoring function to determine the possibility of establishing the triple.

Based on TransE, some methods adjust the framework and structure to integrate experts' prior knowledge of the KG into the model and have achieved even better results in this task. Wang *et al.* [1] improved TransE to TransH by analyzing relations between entities that are reflexive, one-to-many, and many-to-one. Considering that entities always show different behaviors in different aspects, Lin *et al.* [21] proposed TransR, which allows the model to distinguish such properties. The authors mapped model entities and relationships in different spaces and performed transformations in the corresponding relational spaces. Some

further improved models also exist. For example, TransD [176] considers the diversity of entities and relationship diversity, and TransSparse [196] considers the heterogeneity of relationships and the imbalance of the number of head and tail entities.

3) The tensor factorization method represents the semantic features of KGs through tensors, providing better interpretability than deep learning methods. The main idea of tensor factorization is to represent a KG as a three-dimensional tensor and describe the relation prediction as a three-dimensional tensor completion problem. However, the expressiveness of the model is a bottleneck that must be improved. Moreover, enhancing expressiveness tends to increase the complexity of the model. Therefore, balancing expressiveness and complexity remains an open challenge.

RESCAL [177] models the relation matrix as a full-rank matrix, making it a highly expressive model. However, the large number of parameters makes the model prone to overfitting. At the same time, the number of relation representation parameters increases rapidly with the number of relations in the KG, which is not suitable for use in large-scale KGs. Trouillon *et al.* [178] used complex-valued embedding to reconcile expressiveness and complexity. According to the authors, the traditional embedding method can be simplified in the complex number domain, which greatly improves the efficiency of the algorithm. Balažević *et al.* [172] proposed TuckER, which linearly represents each relationship representation by storing part of the knowledge information in a core tensor. This technique reduces the number of parameters while ensuring the expressiveness of the model.

Although significant work has been conducted in this domain, there exist some bottlenecks. For instance, most studies have used general datasets (WN18, FB15K, etc.) only. Corresponding priors may not be applicable to all KGs, especially those in professional domains. The results obtained in this manner cannot ensure KG accuracy, and the predicted results cannot be used directly. Even combination with other automated methods will aggravate the phenomenon of semantic drift, which is still a problem to be solved in relation prediction of KGs.

Additionally, it is pertinent to note that most studies have only predicted the relationship between existing entities in a KG. These studies failed to predict entities and their corresponding relations that do not appear in the KG. Therefore, these methods do not present full advantages in domains where rapid changes occur and new entities continue to appear. The open-world KGC task [179], a new type of KG construction task, aims to address this issue. This task relaxes the assumption of the closed-world hypothesis, enabling the KG to adapt to an environment in which entities are rapidly developing. However, this new direction requires further research.

4.3.2. Completion of Entity Types

The task of completion of entity types is generally called entity typing, which aims to infer the missing type

of an entity. There exist statistical and learning-based methods for entity typing.

1) Entity typing through a heuristic probability model is a traditional statistical method. For example, Paulheim *et al.* [180] proposed SDtype, which uses a type inference mechanism based on heuristic links. The model infers the type of an entity by counting the types of possible entities that have a higher connection frequency with all relationships connected to the entity. Because the probabilistic method used by SDtype ignores the specificity of knowledge expression, SDtype encounters difficulties in fine-grained classes. There are bottlenecks in SDtype's effectiveness, although it is good at predicting classes with a higher frequency and can make large-scale corrections.

2) Learning-based methods for entity typing train a classifier to predict the type for entities. Earlier prediction methods made predictions by manually mining the entity representation patterns of a KG. In recent years, the use of DNNs for representation learning has gradually become a common choice because of the increasing requirements for model performance capabilities [197]. These entity typing methods map entities and relationships to a low-dimensional feature space through a DNN [198].

4.3.3. Completion of Attributes

Completion of attributes is a complex task compared to entity typing and relation prediction. First, it is often difficult to notice the existence of missing attributes because the lack of attributes of entities is often implicit, which is different from entity typing. Second, after discovering the missing attributes, the classification method may be difficult to apply widely due to the different hierarchical structures of different entities. Finally, the completion of numerical attributes may require the introduction of regression methods under some accuracy requirements. However, the effect may not be strong because deviations beyond the given accuracy are unbearable.

Razniewski *et al.* [181] suggested that the lack of attributes of an entity can be judged 1) by referring to similar entities, 2) according to the amount of access of the entity, 3) according to the growth pattern of an entity's attributes, 4) through the pattern matching method, etc. Liang *et al.* proposed Probase+ [182], which uses collaborative filtering. It assumes that similar entities have similar upper and lower levels. A large number of hypernyms and hyponyms in a KG helps this method to find missing attributes for entities.

After attributes have been identified as missing, external completion approaches, such as crowdsourcing [62] and distant supervised [95] methods, can be deployed to complete the attributes.

4.4. Consistency Enhancement of KG

To improve KG consistency, it is common practice to detect and correct inconsistent knowledge. Many studies

have used disjointness axioms to detect inconsistent knowledge in a KG. Yanfang Ma *et al.* [184] analyzed the existing method of discovering disjointness axioms through association rules and proposed an improved method based on the class hierarchy. The authors applied the learned inconsistency axioms to a database, thereby improving the consistency and accuracy of the KG. Another source of additional inconsistency axioms is the use of top-level ontologies, such as *DOLCE* [183], which provides advanced inconsistency axioms [160].

Inconsistent knowledge detection can be performed using knowledge reasoning. However, many KGs come with ontologies that are not sufficiently rich to perform reasoning, such as DBpedia. Therefore, approaches exploiting reasoning are typically used in conjunction with methods for enriching ontologies. For example, Jang *et al.* [185] proposed a semi-automatic quality assessment method. This method analyzes and generates patterns, which can be used to expand the ontology. Tpper *et al.* [186] proposed a method based on an improved DBpedia ontology to identify inconsistencies in the DBpedia dataset. This method uses a rich ontology in the extraction process, which may lead to contradictions pointing to incorrect facts. The authors enriched the ontology, which could then be used to detect inconsistencies. In addition, some methods improve accuracy by boosting consistency [35, 161–165].

4.5. Availability Enhancement of KG

Availability is generally linked with large-scale KGs, as a demand of users, such as in DBpedia and Freebase. The success of these KGs is inseparable from their good accessibility. The subgraphs of interest in large-scale KGs can be accessed in a short time from millions of entities without downloading the entire KG. Query languages, such as SPARQL, can perform a search with detailed constraints [187].

The availability enhancement of large-scale KGs has received research attention recently. To improve the query effect, Verborgh *et al.* [188] implemented a web KG interface to degrade the server load with SPARQL to distribute the load of query evaluation and balance the trade-off between servers and clients. Some studies have focused on the optimization of query results [189, 190]. These studies used example-based methods, such as helping novice users fetch results without expressing their requirements in the form of queries [192]. Lissandrini *et al.* [193] aimed to assist users by expanding their query with intuitive suggestions, which can provide more informative queries to retrieve more detailed and relevant answers. The authors proposed a model that expands information retrieval (IR) methods based on language models and pseudo-relevance feedback, which is described by query-answer graphs through neighboring edge labels. To enhance the query ability of SPARQL, Hu *et al.* [191] proposed a framework called scalable aggregate keyword query over KG (SAKQ), which is able to pose aggregate queries. The authors built a type-predicate graph and constructed query graphs with aggre-

gation, followed by the transformation of query graphs into SPARQL statements with aggregation.

4.6. Trustworthiness Enhancement of KG

Trustworthiness enhancement of KGs is a multifaceted process because trustworthiness is related to various aspects of KG quality [14]. To sort the aspects of the trustworthiness enhancement of KG, we begin with an assessment of the trustworthiness of knowledge (online information) because a KG is usually constructed from online information. The existing assessments of the trustworthiness of online information may be analyzed from three aspects [194]: 1) behavior features related to the source that generates and diffuses information, 2) content-based features related to the textual content of information itself, and 3) graph-based features related to relation networks among information spreaders. The trustworthiness enhancement is analyzed below with respect to these aspects.

1) To enhance the trustworthiness of the knowledge source, knowledge fusion methods¹⁶ should be utilized to fuse a high-quality knowledge base with a KG, which is conducted after the KG has been constructed [152]. In general, well-known open KGs, such as DBpedia and Freebase, retain good trustworthiness and are suitable for knowledge fusion with a KG [38, 61]. The knowledge source selection methods introduced previously could also be employed to choose a high-quality knowledge base to fuse [52, 54]. The correct linkage of entities and relations during knowledge fusion expands a KG with knowledge sources of high quality, thus enhancing the trustworthiness of the knowledge source after KG construction [146, 151].

2) To improve the trustworthiness of information contained in a KG, internal enhancement methods¹⁷ and knowledge fusion methods can be used. The enhancement of other dimensions of KG quality could improve the trustworthiness of KG information, including enhancement in accuracy, timeliness, completeness, and consistency of a KG [14, 44]. Entity linking and knowledge base merging methods can help build expansion information correctly from external knowledge sources with good trustworthiness.

3) Regarding KG spreading routes, they depend on the users who request the KG, a process that remains uncontrolled. However, they can be utilized to support the discovery of information with good trustworthiness. Such analysis of a social network of users for external information could be applied in the selection of high-quality knowledge sources to employ knowledge fusion [195].

5. KG Support for AI Technology

In recent years, big data and AI technologies have been widely used in various fields. Furthermore, KGs and their

robust semantic technologies well suited the provision of assistance in AI technology in many aspects. However, challenges exist in adaptation, as discussed in [199]. KG, as a means of information representation that highlights the relationship between concepts, can solve the problem of AI inexplicability [200]. This section takes the fields of image processing and question answering systems as examples to discuss the support of KGs for AI methods and their requirements for the quality of KGs.

An important means for KGs to support AI algorithms is to use the hierarchical representation and relational structure of KGs to improve the overall understandability of complex AI algorithms [201]. Currently, this type of method has become a research hotspot in many fields [202–204]. In the field of computer vision, many methods introduce this explicit semantic relationship representation into image processing tasks to reduce the need for training data and improve the comprehensibility of the model [205–207]. For example, Wang *et al.* [208] used a category-semantic KG to train a visual classifier, which embeds the image semantics and KG to display the relationship information to train a graph convolutional network and obtain the visual classifier of the new category. Xu *et al.* [209] used a category-to-category KG to solve the problem of semantic dependence between objects and developed a general model with adaptive global reasoning to achieve large-scale target detection. Zhang *et al.* [210] adopted the transformer model for an image captioning task. They took the pretrained bottom-up attention features of images as the input of the encoding. By leveraging the KG, they replaced the word embedding features with the neighborhood words as the representation so that the model could leverage the information of related words during training. Tiwari *et al.* [211] proposed a shared learning approach to learn semantic attributes of images by combining a KG embedding model with the recognized attributes of images. The presented knowledge representation model solves the link prediction task between images.

Generally, for AI technologies supported by KGs, most tasks combine KGs with deep learning methods, where embedded representations of the graphs are required [212, 213], which have higher requirements for the accuracy of KGs. For image processing tasks, most existing methods use the structural information of the relationship between the concepts of KGs. The rupture of the relationship and lack of entities in a KG affect the learning effectiveness of the model, so related technologies are highly sensitive to the completeness of KGs.

For question answering systems, knowledge-based question answering (KBQA) has become an active research direction in recent years. Many deep learning-based methods have been proposed. These methods usually encode questions and candidate answers into distributed representations and select the most appropriate answer by designing different similarity calculations [214]. These methods require a KG to have 1) good logical consistency and 2) good semantic accuracy of relations and structural accuracy of

¹⁶The detailed methods for knowledge fusion are introduced in Section 3.3.

¹⁷Introduced in Sections 4.1, 4.3, and 4.4.

the KG. A KG provides a semantic structure of knowledge through the relation connections between entities, which can be used to infer indirect facts. These characteristics of KGs are extremely beneficial for question and answering systems. The semantic structure of KGs is the basis of reasoning, which requires good semantic and structural accuracy. In addition, a good question answering system should not give contradicting answers to the same question, which requires good consistency of KGs. For example, Huang *et al.* [215] proposed a method that learns the head entity, predicate, and tail entity of a question through KG embedding. They matched the predicate in the question with relations in KGs and inferred the answer through the similarity of vector representations. The semantic matching of predicates in the question and relations in the KG requires the semantic accuracy of relations. Zhang *et al.* [216] proposed an end-to-end variational reasoning network for voice question answering systems. This model infers topic entities in sound through a classifier and completes the answer through multi-hop reasoning using a KG embedding. Good embedding representations require good graph structures, which refer to the accuracy of the graph structure of KGs.

6. Conclusion

KGs are widely used in various applications and have received increasing attention in recent years. The quality evaluation of KGs plays an important role in the application of KGs. However, a comprehensive survey on the quality of KGs has not previously been conducted. This work aimed to fill this gap and facilitate future research in this domain. Six dimensions of KG quality were explored in this survey. This paper introduced quality control in the KG construction process and elucidated the importance of KG quality in KG-based applications. This paper also discussed the quality enhancement methods for KG construction. The extensive review of the literature and the detailed analysis presented in this survey paper could be used to gain insights on the topic as well as to draw future research directions in this area.

Declaration of Competing Interest

The authors declare that they have no conflict of interest in this work.

Acknowledgement

This project is sponsored by NSFC grants 62176245 and 62137002, the Fundamental Research Funds for the Central Universities, Anhui Province grants 202104a05020011 and 202103a07020002.

References

References

- [1] Z. Wang, J. Zhang, J. Feng, Z. Chen, Knowledge graph embedding by translating on hyperplanes, in: Proceedings of the AAAI Conference on Artificial Intelligence, volume 28, 2014.
- [2] H. Paulheim, Knowledge graph refinement: A survey of approaches and evaluation methods, *Semantic web* 8 (2017) 489–508.
- [3] M. Kroetsch, G. Weikum, Special issue on knowledge graphs, *Journal of Web Semantics* 37 (2016) 53–54.
- [4] M. Färber, A. Rettinger, Which knowledge graph is best for me?, *arXiv preprint arXiv:1809.11099* (2018).
- [5] L. Ehrlinger, W. Wöß, Towards a definition of knowledge graphs., *SEMANTiCS (Posters, Demos, SuCCESS)* 48 (2016) 2.
- [6] Y. Duan, L. Shao, G. Hu, Z. Zhou, Q. Zou, Z. Lin, Specifying architecture of knowledge graph with data graph, information graph, knowledge graph and wisdom graph (2017) 327–332.
- [7] C. Xiong, R. Power, J. Callan, Explicit semantic ranking for academic search via knowledge graph embedding, in: Proceedings of the 26th international conference on world wide web, 2017, pp. 1271–1279.
- [8] J. R. Pérez-Agüera, J. Arroyo, J. Greenberg, J. P. Iglesias, V. Fresno, Using bm25f for semantic search, in: Proceedings of the 3rd international semantic search workshop, 2010, pp. 1–8.
- [9] S. Ji, S. Pan, E. Cambria, P. Marttinen, S. Y. Philip, A survey on knowledge graphs: Representation, acquisition, and applications, *IEEE Transactions on Neural Networks and Learning Systems* (2021).
- [10] W. Zhang, S. Deng, H. Wang, Q. Chen, W. Zhang, H. Chen, Xtranse: Explainable knowledge graph embedding for link prediction with lifestyles in e-commerce, in: Joint International Semantic Technology Conference, Springer, 2019, pp. 78–87.
- [11] F.-L. Li, H. Chen, G. Xu, T. Qiu, F. Ji, J. Zhang, H. Chen, Alimekg: Domain knowledge graph construction and application in e-commerce, in: Proceedings of the 29th ACM International Conference on Information & Knowledge Management, 2020, pp. 2581–2588.
- [12] X. Xiang, Z. Wang, Y. Jia, B. Fang, Knowledge graph-based clinical decision support system reasoning: A survey, in: 2019 IEEE Fourth International Conference on Data Science in Cyberspace (DSC), IEEE, 2019, pp. 373–380.
- [13] R. Lourdasamy, X. J. Mattam, Resource description framework based semantic knowledge graph for clinical decision support systems (2021) 69–86.
- [14] A. Zaveri, A. Rula, A. Maurino, R. Pietrobon, J. Lehmann, S. Auer, Quality assessment for linked data: A survey, *Semantic Web* 7 (2016) 63–93.
- [15] R. Y. Wang, D. M. Strong, Beyond accuracy: What data quality means to data consumers, *Journal of management information systems* 12 (1996) 5–33.
- [16] S. Issa, O. Adekunle, F. Hamdi, S. S.-S. Cherfi, M. Dumontier, A. Zaveri, Knowledge graph completeness: A systematic literature review, *IEEE Access* 9 (2021) 31322–31339.
- [17] L. LiuQiao, L. DuanHong, et al., Knowledge graph construction techniques, *Journal of computer research and development* 53 (2016) 582.
- [18] Z. Li, H. Li, H. Wang, Y. Yang, X. Zhang, X. Zhou, Overcoming semantic drift in information extraction., in: 17th International Conference on Extending Database Technology, EDBT 2014, Citeseer, 2014, pp. 169–180.
- [19] V. Yadav, R. Sharp, S. Bethard, Deep affix features improve neural named entity recognizers, in: Proceedings of the seventh joint conference on lexical and computational semantics, 2018, pp. 167–172.
- [20] M. Miwa, M. Bansal, End-to-end relation extraction using lstms on sequences and tree structures, *arXiv preprint arXiv:1601.00770* (2016).

- [21] Y. Lin, Z. Liu, M. Sun, Y. Liu, X. Zhu, Learning entity and relation embeddings for knowledge graph completion, in: Proceedings of the AAAI Conference on Artificial Intelligence, volume 29, 2015.
- [22] J. Liang, S. Zhang, Y. Xiao, How to keep a knowledge base synchronized with its encyclopedia source, in: Proceedings of the Twenty-Sixth International Joint Conference on Artificial Intelligence, IJCAI-17, 2017, pp. 3749–3755. URL: <https://doi.org/10.24963/ijcai.2017/524>. doi:10.24963/ijcai.2017/524.
- [23] A. Melo, H. Paulheim, Detection of relation assertion errors in knowledge graphs, in: Proceedings of the Knowledge Capture Conference, 2017, pp. 1–8.
- [24] A. Zaveri, A. Rula, A. Maurino, R. Pietrobon, J. Lehmann, S. Auer, P. Hitzler, Quality assessment methodologies for linked open data, Submitted to Semantic Web Journal 1 (2013) 1–5.
- [25] B. Stvilia, L. Gasser, M. B. Twidale, L. C. Smith, A framework for information quality assessment, Journal of the American society for information science and technology 58 (2007) 1720–1733.
- [26] R. Zhang, M. Indulska, S. Sadiq, Discovering data quality problems, Business & Information Systems Engineering 61 (2019) 575–593.
- [27] C. Fürber, M. Hepp, Swiqa—a semantic web information quality assessment framework, ECIS 2011 Proceedings (2011).
- [28] F. Naumann, Quality-driven query answering for integrated information systems, volume 2261, Springer, 2003.
- [29] A. Hogan, A. Harth, A. Passant, S. Decker, A. Polleres, Weaving the pedantic web., LDOW 628 (2010) 26.
- [30] Y. Gil, D. Artz, Towards content trust of web resources, Journal of Web Semantics 5 (2007) 227–239.
- [31] C. Bizer, Quality-driven information filtering in the context of web-based information systems, Ph.D. thesis, 2007.
- [32] A. Flemming, Qualitätsmerkmale von Linked Data-veröffentlichenden Datenquellen, Ph.D. thesis, Diplomarbeit (Quality Criteria for Linked Data Sources), 2011.
- [33] Y. Lei, V. Uren, E. Motta, A framework for evaluating semantic metadata, in: Proceedings of the 4th international conference on Knowledge capture, 2007, pp. 135–142.
- [34] A. Zaveri, D. Kontokostas, M. A. Sherif, L. Bühmann, M. Morsey, S. Auer, J. Lehmann, User-driven quality evaluation of dbpedia, in: Proceedings of the 9th International Conference on Semantic Systems, 2013, pp. 97–104.
- [35] H. Li, Y. Li, F. Xu, X. Zhong, Probabilistic error detecting in numerical linked data, in: Database and Expert Systems Applications, Springer, 2015, pp. 61–75.
- [36] P. N. Mendes, H. Mühleisen, C. Bizer, Sieve: linked data quality assessment and fusion, in: Proceedings of the 2012 Joint EDBT/ICDT Workshops, 2012, pp. 116–123.
- [37] M. Luggen, D. Difallah, C. Sarasua, G. Demartini, P. Cudré-Mauroux, Non-parametric class completeness estimators for collaborative knowledge graphs: the case of wikidata, in: International Semantic Web Conference, Springer, 2019, pp. 453–469.
- [38] M. Färber, F. Bartscherer, C. Menne, A. Rettinger, Linked data quality of dbpedia, freebase, opencyc, wikidata, and yago, Semantic Web 9 (2018) 77–129.
- [39] V. Jayawardene, S. Sadiq, M. Indulska, An analysis of data quality dimensions, ITEE Technical Report (2015) 1–32.
- [40] M. Gamble, C. Goble, Quality, trust, and utility of scientific data on the web: Towards a joint model, in: Proceedings of the 3rd international web science conference, 2011, pp. 1–8.
- [41] C. Bizer, J. Lehmann, G. Kobilarov, S. Auer, C. Becker, R. Cyganiak, S. Hellmann, Dbpedia—a crystallization point for the web of data, Journal of web semantics 7 (2009) 154–165.
- [42] C. Guéret, P. Groth, C. Stadler, J. Lehmann, Assessing linked data mappings using network measures, in: Extended semantic web conference, Springer, 2012, pp. 87–102.
- [43] F. Bray, D. M. Parkin, Evaluation of data quality in the cancer registry: principles and methods. part i: comparability, validity and timeliness, European journal of cancer 45 (2009) 747–755.
- [44] U. Dayton, B. Mons, H. Paulheim, Knowledge graphs: New directions for knowledge representation on the semantic web, Knowledge Dynamics and Evolution—No Dynamic or Evolving Knowledge Graph Without Provenance (2018) 58.
- [45] G. K. Tayi, D. P. Ballou, Examining data quality, Communications of the ACM 41 (1998) 54–57.
- [46] N. Mihindukulasooriya, R. García-Castro, A. Gómez-Pérez, Ld sniffer: A quality assessment tool for measuring the accessibility of linked data, in: European Knowledge Acquisition Workshop, Springer, 2016, pp. 149–152.
- [47] B. J. Fogg, C. Soohoo, D. R. Danielson, L. Marable, J. Stanford, E. R. Tauber, How do users evaluate the credibility of web sites? a study with over 2,500 participants, in: Proceedings of the 2003 conference on Designing for user experiences, 2003, pp. 1–15.
- [48] A. N. Langville, C. D. Meyer, Deeper inside pagerank, Internet Mathematics 1 (2004) 335–380.
- [49] L. Page, S. Brin, R. Motwani, T. Winograd, The PageRank citation ranking: Bringing order to the web., Technical Report, Stanford InfoLab, 1999.
- [50] J. M. Kleinberg, Authoritative sources in a hyperlinked environment, Journal of the ACM (JACM) 46 (1999) 604–632.
- [51] G. A. Wang, J. Jiao, A. S. Abrahams, W. Fan, Z. Zhang, Expertrank: A topic-aware expert finding algorithm for online knowledge communities, Decision support systems 54 (2013) 1442–1451.
- [52] P. Nguyen, P. Tomeo, T. Di Noia, E. Di Sciascio, An evaluation of simrank and personalized pagerank to build a recommender system for the web of data, in: Proceedings of the 24th International Conference on World Wide Web, 2015, pp. 1477–1482.
- [53] R. Hao, J. Huang, On construction method of shipborne and airborne radar intelligence and related equipment knowledge graph, in: Journal of Physics: Conference Series, volume 887, IOP Publishing, 2017, p. 012042.
- [54] K. Popat, S. Mukherjee, J. Strötgen, G. Weikum, Credibility assessment of textual claims on the web, in: Proceedings of the 25th ACM International on Conference on Information and Knowledge Management, 2016, pp. 2173–2178.
- [55] A. Olteanu, S. Peshterliev, X. Liu, K. Aberer, Web credibility: Features exploration and credibility prediction, in: European conference on information retrieval, Springer, 2013, pp. 557–568.
- [56] L. Yi, B. Liu, X. Li, Eliminating noisy information in web pages for data mining, in: Proceedings of the ninth ACM SIGKDD international conference on Knowledge discovery and data mining, 2003, pp. 296–305.
- [57] D. Gibson, K. Punera, A. Tomkins, The volume and evolution of web page templates, in: Special interest tracks and posters of the 14th international conference on World Wide Web, 2005, pp. 830–839.
- [58] W. Liu, X. Meng, W. Meng, Vision-based web data records extraction., in: WebDB, 2006.
- [59] S. Debnath, P. Mitra, N. Pal, C. L. Giles, Automatic identification of informative sections of web pages, IEEE transactions on knowledge and data engineering 17 (2005) 1233–1246.
- [60] D. Vrandečić, M. Krötzsch, Wikidata: a free collaborative knowledgebase, Communications of the ACM 57 (2014) 78–85.
- [61] K. Bollacker, C. Evans, P. Paritosh, T. Sturge, J. Taylor, Freebase: a collaboratively created graph database for structuring human knowledge, in: Proceedings of the 2008 ACM SIGMOD international conference on Management of data, 2008, pp. 1247–1250.
- [62] M. Lease, On quality control and machine learning in crowdsourcing., Human Computation 11 (2011).
- [63] A. P. Dawid, A. M. Skene, Maximum likelihood estimation of observer error-rates using the em algorithm, Journal of the Royal Statistical Society: Series C (Applied Statistics) 28

- (1979) 20–28.
- [64] P. G. Ipeirotis, F. Provost, J. Wang, Quality management on amazon mechanical turk, in: *Proceedings of the ACM SIGKDD workshop on human computation*, 2010, pp. 64–67.
- [65] V. C. Raykar, S. Yu, Eliminating spammers and ranking annotators for crowdsourced labeling tasks, *The Journal of Machine Learning Research* 13 (2012) 491–518.
- [66] H. Chen, X. Yao, Multiobjective neural network ensembles based on regularized negative correlation learning, *IEEE Transactions on Knowledge and Data Engineering* 22 (2010) 1738–1751.
- [67] H. Chen, P. Tino, X. Yao, Probabilistic classification vector machines, *IEEE Transactions on Neural Networks* 20 (2009) 901–914.
- [68] H. Chen, P. Tiño, X. Yao, Predictive ensemble pruning by expectation propagation, *IEEE Transactions on Knowledge and Data Engineering* 21 (2009) 999–1013.
- [69] H. Chen, P. Tiño, A. Rodan, X. Yao, Learning in the model space for cognitive fault diagnosis, *IEEE transactions on neural networks and learning systems* 25 (2013) 124–136.
- [70] H. Chen, F. Tang, P. Tino, X. Yao, Model-based kernel for efficient time series analysis, in: *Proceedings of the 19th ACM SIGKDD international conference on Knowledge discovery and data mining*, 2013, pp. 392–400.
- [71] H. Chen, P. Tiño, X. Yao, Efficient probabilistic classification vector machine with incremental basis function selection, *IEEE transactions on neural networks and learning systems* 25 (2013) 356–369.
- [72] H. Chen, F. Tang, P. Tino, A. G. Cohn, X. Yao, Model metric co-learning for time series classification., in: *Proceedings of the Twenty-Fourth International Joint Conference on Artificial Intelligence*, AAAI Press, 2015, pp. 3387–3394.
- [73] Z. Gong, H. Chen, Model-based oversampling for imbalanced sequence classification, in: *Proceedings of the 25th ACM International on Conference on Information and Knowledge Management*, 2016, pp. 1009–1018.
- [74] A. T. Nguyen, B. C. Wallace, J. J. Li, A. Nenkova, M. Lease, Aggregating and predicting sequence labels from crowd annotations, in: *Proceedings of the conference. Association for Computational Linguistics. Meeting*, volume 2017, NIH Public Access, 2017, p. 299.
- [75] Y. Yang, M. Zhang, W. Chen, W. Zhang, H. Wang, M. Zhang, Adversarial learning for chinese ner from crowd annotations, in: *Proceedings of the AAAI Conference on Artificial Intelligence*, volume 32, 2018.
- [76] C. Ru, J. Tang, S. Li, S. Xie, T. Wang, Using semantic similarity to reduce wrong labels in distant supervision for relation extraction, *Information Processing & Management* 54 (2018) 593–608.
- [77] L. Chiticariu, R. Krishnamurthy, Y. Li, F. Reiss, S. Vaithyanathan, Domain adaptation of rule-based annotators for named-entity recognition tasks, in: *Proceedings of the 2010 conference on empirical methods in natural language processing*, 2010, pp. 1002–1012.
- [78] E. Riloff, R. Jones, et al., Learning dictionaries for information extraction by multi-level bootstrapping, in: *AAAI/IAAI*, 1999, pp. 474–479.
- [79] M. Thelen, E. Riloff, A bootstrapping method for learning semantic lexicons using extraction pattern contexts, in: *Proceedings of the 2002 conference on empirical methods in natural language processing (EMNLP 2002)*, 2002, pp. 214–221.
- [80] P. Pantel, M. Pennacchiotti, Espresso: Leveraging generic patterns for automatically harvesting semantic relations, in: *Proceedings of the 21st International Conference on Computational Linguistics and 44th Annual Meeting of the Association for Computational Linguistics*, 2006, pp. 113–120.
- [81] M. Komachi, T. Kudo, M. Shimbo, Y. Matsumoto, Graph-based analysis of semantic drift in espresso-like bootstrapping algorithms, in: *Proceedings of the 2008 Conference on Empirical Methods in Natural Language Processing*, 2008, pp. 1011–1020.
- [82] M. Pasca, D. Lin, J. Bigham, A. Lifchits, A. Jain, Names and similarities on the web: Fact extraction in the fast lane, *Proceedings of the 21st International Conference on Computational Linguistics and 44th Annual Meeting of the Association for Computational Linguistics* (2006).
- [83] F. Fouss, A. Pirotte, J.-M. Renders, M. Saerens, Random-walk computation of similarities between nodes of a graph with application to collaborative recommendation, *IEEE Transactions on knowledge and data engineering* 19 (2007) 355–369.
- [84] J. R. Curran, T. Murphy, B. Scholz, Minimising semantic drift with mutual exclusion bootstrapping, in: *Proceedings of the 10th Conference of the Pacific Association for Computational Linguistics*, volume 6, Citeseer, 2007, pp. 172–180.
- [85] S. Wang, S. Schlobach, M. Klein, Concept drift and how to identify it, *Journal of Web Semantics* 9 (2011) 247–265.
- [86] T. G. Stavropoulos, S. Andreadis, E. Kontopoulos, I. Kompatsiaris, Semadrift: A hybrid method and visual tools to measure semantic drift in ontologies, *Journal of Web Semantics* 54 (2019) 87–106.
- [87] T. G. Stavropoulos, S. Andreadis, M. Riga, E. Kontopoulos, P. Mitzias, I. Kompatsiaris, A framework for measuring semantic drift in ontologies., in: *SEMANTICS (Posters, Demos, SuCCESS)*, 2016.
- [88] A. Meroño-Peñuela, R. Hoekstra, What is linked historical data?, in: *International Conference on Knowledge Engineering and Knowledge Management*, Springer, 2014, pp. 282–287.
- [89] G. Zhou, J. Su, Named entity recognition using an hmm-based chunk tagger, in: *Proceedings of the 40th Annual Meeting of the Association for Computational Linguistics*, 2002, pp. 473–480.
- [90] X. Ma, E. Hovy, End-to-end sequence labeling via bi-directional lstm-cnns-crf, *arXiv preprint arXiv:1603.01354* (2016).
- [91] R. Malouf, Markov models for language-independent named entity recognition, in: *COLING-02: The 6th Conference on Natural Language Learning 2002 (CoNLL-2002)*, 2002.
- [92] Y. Li, K. Bontcheva, H. Cunningham, Svm based learning system for information extraction, in: *International Workshop on Deterministic and Statistical Methods in Machine Learning*, Springer, 2004, pp. 319–339.
- [93] S. Li, J. Graça, B. Taskar, Wiki-ly supervised part-of-speech tagging, in: *Proceedings of the 2012 Joint Conference on Empirical Methods in Natural Language Processing and Computational Natural Language Learning*, 2012, pp. 1389–1398.
- [94] T. Rocktäschel, T. Huber, M. Weidlich, U. Leser, Wbi-ner: The impact of domain-specific features on the performance of identifying and classifying mentions of drugs, in: *Second Joint Conference on Lexical and Computational Semantics (*SEM)*, Volume 2: *Proceedings of the Seventh International Workshop on Semantic Evaluation (SemEval 2013)*, 2013, pp. 356–363.
- [95] M. Fang, T. Cohn, Learning when to trust distant supervision: An application to low-resource pos tagging using cross-lingual projection, *arXiv preprint arXiv:1607.01133* (2016).
- [96] S. Liu, B. Tang, Q. Chen, X. Wang, Effects of semantic features on machine learning-based drug name recognition systems: word embeddings vs. manually constructed dictionaries, *Information* 6 (2015) 848–865.
- [97] Y. Lin, S. Shen, Z. Liu, H. Luan, M. Sun, Neural relation extraction with selective attention over instances, in: *Proceedings of the 54th Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, 2016, pp. 2124–2133.
- [98] Y. Bengio, P. Simard, P. Frasconi, Learning long-term dependencies with gradient descent is difficult, *IEEE transactions on neural networks* 5 (1994) 157–166.
- [99] A. Ratner, S. H. Bach, H. Ehrenberg, J. Fries, S. Wu, C. Ré, Snorkel: Rapid training data creation with weak supervision, in: *Proceedings of the VLDB Endowment. International Conference on Very Large Data Bases*, volume 11, NIH Public Access, 2017, p. 269.
- [100] Z. Huang, W. Xu, K. Yu, Bidirectional lstm-crf models for

- sequence tagging, arXiv preprint arXiv:1508.01991 (2015).
- [101] C. Jia, Y. Zhang, Multi-cell compositional lstm for ner domain adaptation, in: Proceedings of the 58th Annual Meeting of the Association for Computational Linguistics, 2020, pp. 5906–5917.
 - [102] J. T. Zhou, H. Zhang, D. Jin, H. Zhu, M. Fang, R. S. M. Goh, K. Kwok, Dual adversarial neural transfer for low-resource named entity recognition, in: Proceedings of the 57th Annual Meeting of the Association for Computational Linguistics, 2019, pp. 3461–3471.
 - [103] C. Jia, X. Liang, Y. Zhang, Cross-domain ner using cross-domain language modeling, in: Proceedings of the 57th Annual Meeting of the Association for Computational Linguistics, 2019, pp. 2464–2474.
 - [104] J. D. Rodriguez, A. Caldwell, A. Liu, Transfer learning for entity recognition of novel classes, in: Proceedings of the 27th International Conference on Computational Linguistics, 2018, pp. 1974–1985.
 - [105] P. Cao, Y. Chen, K. Liu, J. Zhao, S. Liu, Adversarial transfer learning for chinese named entity recognition with self-attention mechanism, in: Proceedings of the 2018 Conference on Empirical Methods in Natural Language Processing, 2018, pp. 182–192.
 - [106] H. Yang, S. Huang, X. Dai, J. Chen, Fine-grained knowledge fusion for sequence labeling domain adaptation, arXiv preprint arXiv:1909.04315 (2019).
 - [107] Y. Yang, W. Chen, Z. Li, Z. He, M. Zhang, Distantly supervised ner with partial annotation learning and reinforcement learning, in: Proceedings of the 27th International Conference on Computational Linguistics, 2018, pp. 2159–2169.
 - [108] J. Shang, L. Liu, X. Ren, X. Gu, T. Ren, J. Han, Learning named entity tagger using domain-specific dictionary, arXiv preprint arXiv:1809.03599 (2018).
 - [109] J. Ni, G. Dinu, R. Florian, Weakly supervised cross-lingual named entity recognition via effective annotation and representation projection, arXiv preprint arXiv:1707.02483 (2017).
 - [110] Y. Cao, Z. Hu, T.-S. Chua, Z. Liu, H. Ji, Low-resource name tagging learned with weakly labeled data, arXiv preprint arXiv:1908.09659 (2019).
 - [111] C. Liang, Y. Yu, H. Jiang, S. Er, R. Wang, T. Zhao, C. Zhang, Bond: Bert-assisted open-domain named entity recognition with distant supervision, in: Proceedings of the 26th ACM SIGKDD International Conference on Knowledge Discovery & Data Mining, 2020, pp. 1054–1064.
 - [112] S. Brin, Extracting patterns and relations from the world wide web, in: International workshop on the world wide web and databases, Springer, 1998, pp. 172–183.
 - [113] E. Agichtein, L. Gravano, Snowball: Extracting relations from large plain-text collections, in: Proceedings of the fifth ACM conference on Digital libraries, 2000, pp. 85–94.
 - [114] N. Kambhatla, Combining lexical, syntactic, and semantic features with maximum entropy models for information extraction, in: Proceedings of the ACL Interactive Poster and Demonstration Sessions, 2004, pp. 178–181.
 - [115] D. Zeng, K. Liu, Y. Chen, J. Zhao, Distant supervision for relation extraction via piecewise convolutional neural networks, in: Proceedings of the 2015 conference on empirical methods in natural language processing, 2015, pp. 1753–1762.
 - [116] G. Zhou, J. Su, J. Zhang, M. Zhang, Exploring various knowledge in relation extraction, in: Proceedings of the 43rd annual meeting of the association for computational linguistics (acl05), 2005, pp. 427–434.
 - [117] R. Bunescu, R. J. Mooney, Subsequence kernels for relation extraction, in: NIPS, Citeseer, 2005, pp. 171–178.
 - [118] T. Hasegawa, S. Sekine, R. Grishman, Discovering relations among named entities from large corpora, in: Proceedings of the 42nd Annual Meeting of the Association for Computational Linguistics (ACL-04), 2004, pp. 415–422.
 - [119] M. Collins, N. Duffy, Convolution kernels for natural language, in: Advances in neural information processing systems, 2001, pp. 625–632.
 - [120] D. Lin, P. Pantel, Discovery of inference rules for question-answering, *Natural Language Engineering* 7 (2001) 343–360.
 - [121] D. Zeng, K. Liu, S. Lai, G. Zhou, J. Zhao, Relation classification via convolutional deep neural network, in: Proceedings of COLING 2014, the 25th international conference on computational linguistics: technical papers, 2014, pp. 2335–2344.
 - [122] H. Chen, P. Hong, W. Han, N. Majumder, S. Poria, Dialogue relation extraction with document-level heterogeneous graph attention networks, arXiv preprint arXiv:2009.05092 (2020).
 - [123] Y. Hong, Y. Liu, S. Yang, K. Zhang, A. Wen, J. Hu, Improving graph convolutional networks based on relation-aware attention for end-to-end relation extraction, *IEEE Access* 8 (2020) 51315–51323.
 - [124] G. Ji, K. Liu, S. He, J. Zhao, Distant supervision for relation extraction with sentence-level attention and entity descriptions, in: Proceedings of the AAAI Conference on Artificial Intelligence, volume 31, 2017.
 - [125] Z. Geng, G. Chen, Y. Han, G. Lu, F. Li, Semantic relation extraction using sequential and tree-structured lstm with attention, *Information Sciences* 509 (2020) 183–192.
 - [126] Y. Li, G. Long, T. Shen, T. Zhou, L. Yao, H. Huo, J. Jiang, Self-attention enhanced selective gate with entity-aware embedding for distantly supervised relation extraction, in: Proceedings of the AAAI Conference on Artificial Intelligence, volume 34, 2020, pp. 8269–8276.
 - [127] H. Fei, Y. Ren, D. Ji, Boundaries and edges rethinking: An end-to-end neural model for overlapping entity relation extraction, *Information Processing & Management* 57 (2020) 102311.
 - [128] G. Bekoulis, J. Deleu, T. Demeester, C. Develder, Joint entity recognition and relation extraction as a multi-head selection problem, *Expert Systems with Applications* 114 (2018) 34–45.
 - [129] S. Zheng, F. Wang, H. Bao, Y. Hao, P. Zhou, B. Xu, Joint extraction of entities and relations based on a novel tagging scheme, arXiv preprint arXiv:1706.05075 (2017).
 - [130] D. Zeng, H. Zhang, Q. Liu, Copymtl: Copy mechanism for joint extraction of entities and relations with multi-task learning, in: Proceedings of the AAAI Conference on Artificial Intelligence, volume 34, 2020, pp. 9507–9514.
 - [131] X. Zeng, D. Zeng, S. He, K. Liu, J. Zhao, Extracting relational facts by an end-to-end neural model with copy mechanism, in: Proceedings of the 56th Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers), 2018, pp. 506–514.
 - [132] O. Rusu, I. Halcu, O. Grigoriu, G. Neculoiu, V. Sandulescu, M. Marinescu, V. Marinescu, Converting unstructured and semi-structured data into knowledge (2013) 1–4.
 - [133] Z. Nasar, S. W. Jaffry, M. K. Malik, Named entity recognition and relation extraction: State-of-the-art, *ACM Computing Surveys (CSUR)* 54 (2021) 1–39.
 - [134] D. Nadeau, S. Sekine, A survey of named entity recognition and classification, *Linguisticae Investigationes* 30 (2007) 3–26.
 - [135] S. D. Pedro, E. R. Hruschka, Crowd-powered systems to diminish the effects of semantic drift, in: *International Conference on Hybrid Artificial Intelligence Systems*, Springer, 2019, pp. 697–709.
 - [136] S. Pawar, G. K. Palshikar, P. Bhattacharyya, Relation extraction: A survey, arXiv preprint arXiv:1712.05191 (2017).
 - [137] H. L. Nguyen, D. T. Vu, J. J. Jung, Knowledge graph fusion for smart systems: A survey, *Information Fusion* 61 (2020) 56–70.
 - [138] A. Bagga, B. Baldwin, Entity-based cross-document core fereencing using the vector space model, in: 36th Annual Meeting of the Association for Computational Linguistics and 17th International Conference on Computational Linguistics, Volume 1, 1998, pp. 79–85.
 - [139] S. Cucerzan, Large-scale named entity disambiguation based on wikipedia data, in: Proceedings of the 2007 joint conference on empirical methods in natural language processing and computational natural language learning (EMNLP-CoNLL), 2007, pp. 708–716.
 - [140] B. Malin, E. Airoidi, K. M. Carley, A network analysis model

- for disambiguation of names in lists, *Computational & Mathematical Organization Theory* 11 (2005) 119–139.
- [141] S. Lappin, H. J. Leass, An algorithm for pronominal anaphora resolution, *Computational linguistics* 20 (1994) 535–561.
- [142] T. L. Kennedy, Cystic hygroma-lymphangioma: a rare and still unclear entity, *The Laryngoscope* 99 (1989) 1–10.
- [143] P. Besana, A framework for combining ontology and schema matchers with dempster-shafer, in: *Proceedings of the 1st International Conference on Ontology Matching-Volume 225*, 2006, pp. 196–200.
- [144] P. D. Turney, Mining the web for synonyms: Pmi-ir versus lsa on toefl, in: *European conference on machine learning*, Springer, 2001, pp. 491–502.
- [145] Z. He, S. Liu, M. Li, M. Zhou, L. Zhang, H. Wang, Learning entity representation for entity disambiguation, in: *Proceedings of the 51st Annual Meeting of the Association for Computational Linguistics (Volume 2: Short Papers)*, 2013, pp. 30–34.
- [146] N. Kolitsas, O.-E. Ganea, T. Hofmann, End-to-end neural entity linking, *arXiv preprint arXiv:1808.07699* (2018).
- [147] O.-E. Ganea, T. Hofmann, Deep joint entity disambiguation with local neural attention, *arXiv preprint arXiv:1704.04920* (2017).
- [148] P. Le, I. Titov, Improving entity linking by modeling latent relations between mentions, *arXiv preprint arXiv:1804.10637* (2018).
- [149] P. Le, I. Titov, Boosting entity linking performance by leveraging unlabeled documents, *arXiv preprint arXiv:1906.01250* (2019).
- [150] P. Le, I. Titov, Distant learning for entity linking with automatic noise detection, *arXiv preprint arXiv:1905.07189* (2019).
- [151] S. Chen, J. Wang, F. Jiang, C.-Y. Lin, Improving entity linking by modeling latent entity type information, in: *Proceedings of the AAAI Conference on Artificial Intelligence*, volume 34, 2020, pp. 7529–7537.
- [152] X. Zhao, Y. Jia, A. Li, R. Jiang, Y. Song, Multi-source knowledge fusion: a survey, *World Wide Web* 23 (2020) 2567–2592.
- [153] Y. Zhang, L. Liu, S. Fu, F. Zhong, Entity alignment across knowledge graphs based on representative relations selection, in: *2018 5th International Conference on Systems and Informatics (ICSAI)*, IEEE, 2018, pp. 1056–1061.
- [154] B. D. Trisedya, J. Qi, R. Zhang, Entity alignment between knowledge graphs using attribute embeddings, in: *Proceedings of the AAAI Conference on Artificial Intelligence*, volume 33, 2019, pp. 297–304.
- [155] S. Guan, X. Jin, Y. Wang, Y. Jia, H. Shen, Z. Li, X. Cheng, Self-learning and embedding based entity alignment, *Knowledge and Information Systems* 59 (2019) 361–386.
- [156] X. Wang, K. Liu, S. He, S. Liu, Y. Zhang, J. Zhao, Multi-source knowledge bases entity alignment by leveraging semantic tags, *Chinese Journal of Computers* 40 (2017) 701–711.
- [157] N. Lao, T. Mitchell, W. Cohen, Random walk inference and learning in a large scale knowledge base, in: *Proceedings of the 2011 conference on empirical methods in natural language processing*, 2011, pp. 529–539.
- [158] H. Paulheim, C. Bizer, Improving the quality of linked data using statistical distributions, *International Journal on Semantic Web and Information Systems (IJSWIS)* 10 (2014) 63–86.
- [159] X. Dong, E. Gabrilovich, G. Heitz, W. Horn, N. Lao, K. Murphy, T. Strohmann, S. Sun, W. Zhang, Knowledge vault: A web-scale approach to probabilistic knowledge fusion, in: *Proceedings of the 20th ACM SIGKDD international conference on Knowledge discovery and data mining*, 2014, pp. 601–610.
- [160] H. Paulheim, A. Gangemi, Serving dbpedia with dolce—more than just adding a cherry on top, in: *International Semantic Web Conference*, Springer, 2015, pp. 180–196.
- [161] D. Wienand, H. Paulheim, Detecting incorrect numerical data in dbpedia, in: *European Semantic Web Conference*, Springer, 2014, pp. 504–518.
- [162] D. Fleischhacker, H. Paulheim, V. Bryl, J. Völker, C. Bizer, Detecting errors in numerical linked data using cross-checked outlier detection, in: *International semantic web conference*, Springer, 2014, pp. 357–372.
- [163] L. Golab, H. Karloff, F. Korn, A. Saha, D. Srivastava, Sequential dependencies, *Proceedings of the VLDB Endowment* 2 (2009) 574–585.
- [164] N. Koudas, A. Saha, D. Srivastava, S. Venkatasubramanian, Metric functional dependencies, in: *2009 IEEE 25th International Conference on Data Engineering*, IEEE, 2009, pp. 1275–1278.
- [165] G. Fan, W. Fan, F. Geerts, Detecting errors in numeric attributes, in: *International Conference on Web-Age Information Management*, Springer, 2014, pp. 125–137.
- [166] S. Auer, C. Bizer, G. Kobilarov, J. Lehmann, R. Cyganiak, Z. Ives, Dbpedia: A nucleus for a web of open data, in: *The semantic web*, Springer, 2007, pp. 722–735.
- [167] S. Hellmann, C. Stadler, J. Lehmann, S. Auer, Dbpedia live extraction, in: *OTM Confederated International Conferences” On the Move to Meaningful Internet Systems”*, Springer, 2009, pp. 1209–1223.
- [168] R. L. Winkler, An introduction to Bayesian inference and decision / [by] Robert L. Winkler., Series in quantitative methods for decision making, Holt, Rinehart and Winston, New York, 1972.
- [169] M. Pinsky, S. Karlin, An introduction to stochastic modeling, Academic Press; 4th edition (December 24, 2010), 2010.
- [170] J. Cho, H. Garcia-Molina, Estimating frequency of change, *ACM Transactions on Internet Technology (TOIT)* 3 (2003) 256–290.
- [171] T. Jiang, T. Liu, T. Ge, L. Sha, B. Chang, S. Li, Z. Sui, Towards time-aware knowledge graph completion, in: *Proceedings of COLING 2016, the 26th International Conference on Computational Linguistics: Technical Papers*, 2016, pp. 1715–1724.
- [172] I. Balažević, C. Allen, T. M. Hospedales, Tucker: Tensor factorization for knowledge graph completion, *arXiv preprint arXiv:1901.09590* (2019).
- [173] M. Richardson, P. Domingos, Markov logic networks, *Machine learning* 62 (2006) 107–136.
- [174] Y. Chen, D. Z. Wang, Knowledge expansion over probabilistic knowledge bases, in: *Proceedings of the 2014 ACM SIGMOD international conference on Management of data*, 2014, pp. 649–660.
- [175] A. Bordes, N. Usunier, A. Garcia-Duran, J. Weston, O. Yakhnenko, Translating embeddings for modeling multi-relational data, in: *Neural Information Processing Systems (NIPS)*, 2013, pp. 1–9.
- [176] G. Ji, S. He, L. Xu, K. Liu, J. Zhao, Knowledge graph embedding via dynamic mapping matrix, in: *Proceedings of the 53rd annual meeting of the association for computational linguistics and the 7th international joint conference on natural language processing (volume 1: Long papers)*, 2015, pp. 687–696.
- [177] M. Nickel, V. Tresp, H.-P. Kriegel, A three-way model for collective learning on multi-relational data, in: *Icml*, 2011.
- [178] T. Trouillon, C. R. Dance, J. Welbl, S. Riedel, É. Gaussier, G. Bouchard, Knowledge graph completion via complex tensor factorization, *arXiv preprint arXiv:1702.06879* (2017).
- [179] B. Shi, T. Weninger, Open-world knowledge graph completion, in: *Proceedings of the AAAI Conference on Artificial Intelligence*, volume 32, 2018.
- [180] H. Paulheim, C. Bizer, Type inference on noisy rdf data, in: *International semantic web conference*, Springer, 2013, pp. 510–525.
- [181] S. Razniewski, F. Suchanek, W. Nutt, But what do we actually know?, in: *Proceedings of the 5th Workshop on Automated Knowledge Base Construction*, 2016, pp. 40–44.
- [182] J. Liang, Y. Xiao, H. Wang, Y. Zhang, W. Wang, Probbase+: Inferring missing links in conceptual taxonomies, *IEEE Transactions on Knowledge and Data Engineering* 29 (2017) 1281–1295.
- [183] A. Gangemi, N. Guarino, C. Masolo, A. Oltramari, Sweetening wordnet with dolce, *AI magazine* 24 (2003) 13–13.
- [184] Y. Ma, H. Gao, T. Wu, G. Qi, Learning disjointness axioms

- with association rule mining and its application to inconsistency detection of linked data, in: Chinese Semantic Web and Web Science Conference, Springer, 2014, pp. 29–41.
- [185] S. Jang, M. Megawati, J. Choi, M. Y. Yi, Semi-automatic quality assessment of linked data without requiring ontology., in: NLP-DBPEDIA@ ISWC, 2015, pp. 45–55.
- [186] G. Töpper, M. Knuth, H. Sack, Dbpedia ontology enrichment for inconsistency detection, in: Proceedings of the 8th International Conference on Semantic Systems, 2012, pp. 33–40.
- [187] A. Bonifati, W. Martens, T. Timm, An analytical study of large sparql query logs, *The VLDB Journal* 29 (2020) 655–679.
- [188] R. Verborgh, M. Vander Sande, O. Hartig, J. Van Herwegen, L. De Vocht, B. De Meester, G. Haesendonck, P. Colpaert, Triple pattern fragments: a low-cost knowledge graph interface for the web, *Journal of Web Semantics* 37 (2016) 184–206.
- [189] M. Lissandrini, D. Mottin, T. Palpanas, Y. Velegrakis, Multi-example search in rich information graphs, in: 2018 IEEE 34th International Conference on Data Engineering (ICDE), IEEE, 2018, pp. 809–820.
- [190] D. Mottin, M. Lissandrini, Y. Velegrakis, T. Palpanas, Exemplar queries: a new way of searching, *The VLDB Journal* 25 (2016) 741–765.
- [191] X. Hu, J. Duan, D. Dang, Scalable aggregate keyword query over knowledge graph, *Future Generation Computer Systems* 107 (2020) 588–600.
- [192] M. Lissandrini, D. Mottin, T. Palpanas, Y. Velegrakis, Data exploration using example-based methods, *Synthesis Lectures on Data Management* 10 (2018) 1–164.
- [193] M. Lissandrini, D. Mottin, T. Palpanas, Y. Velegrakis, Graph-query suggestions for knowledge graph exploration, in: Proceedings of The Web Conference 2020, 2020, pp. 2549–2555.
- [194] G. Livraga, M. Viviani, Data confidentiality and information credibility in on-line ecosystems, in: Proceedings of the 11th International Conference on Management of Digital EcoSystems, 2019, pp. 191–198.
- [195] C. Shao, G. L. Ciampaglia, O. Varol, K.-C. Yang, A. Flammini, F. Menczer, The spread of low-credibility content by social bots, *Nature communications* 9 (2018) 1–9.
- [196] G. Ji, K. Liu, S. He, J. Zhao, Knowledge graph completion with adaptive sparse transfer matrix, in: Proceedings of the AAAI Conference on Artificial Intelligence, volume 30, 2016.
- [197] S. Shimaoka, P. Stenetorp, K. Inui, S. Riedel, Neural architectures for fine-grained entity type classification, *arXiv preprint arXiv:1606.01341* (2016).
- [198] D. Yogatama, D. Gillick, N. Lazic, Embedding methods for fine grained entity type classification, in: Proceedings of the 53rd Annual Meeting of the Association for Computational Linguistics and the 7th International Joint Conference on Natural Language Processing (Volume 2: Short Papers), 2015, pp. 291–296.
- [199] G. Futia, A. Vetrò, On the integration of knowledge graphs into deep learning models for a more comprehensible ai: three challenges for future research, *Information* 11 (2020) 122.
- [200] F. Lecue, On the role of knowledge graphs in explainable ai, *Semantic Web* 11 (2020) 41–51.
- [201] M. K. Sarker, J. Schwartz, P. Hitzler, L. Zhou, S. Nadella, B. Minnery, I. Juvina, M. L. Raymer, W. R. Aue, Wikipedia knowledge graph for explainable ai, in: Iberoamerican Knowledge Graphs and Semantic Web Conference, Springer, 2020, pp. 72–87.
- [202] Y. Qiu, Y. Wang, X. Jin, K. Zhang, Stepwise reasoning for multi-relation question answering over knowledge graph with weak supervision, in: Proceedings of the 13th International Conference on Web Search and Data Mining, 2020, pp. 474–482.
- [203] W. Ma, M. Zhang, Y. Cao, W. Jin, C. Wang, Y. Liu, S. Ma, X. Ren, Jointly learning explainable rules for recommendation with knowledge graph, in: The World Wide Web Conference, 2019, pp. 1210–1221.
- [204] C. Zhao, C. Xiong, X. Qian, J. Boyd-Graber, Complex factoid question answering with a free-text knowledge graph, in: Proceedings of The Web Conference 2020, 2020, pp. 1205–1216.
- [205] R. Chen, T. Chen, X. Hui, H. Wu, G. Li, L. Lin, Knowledge graph transfer network for few-shot recognition, in: Proceedings of the AAAI Conference on Artificial Intelligence, volume 34, 2020, pp. 10575–10582.
- [206] Y. Zhu, F. Zhuang, X. Zhang, Z. Qi, Z. Shi, Q. He, Combat data shift in few-shot learning with knowledge graph, *arXiv preprint arXiv:2101.11354* (2021).
- [207] Y. Geng, J. Chen, Z. Chen, J. Z. Pan, Z. Yuan, H. Chen, Kzsl: Resources for knowledge-driven zero-shot learning, *arXiv preprint arXiv:2106.15047* (2021).
- [208] X. Wang, Y. Ye, A. Gupta, Zero-shot recognition via semantic embeddings and knowledge graphs, in: Proceedings of the IEEE conference on computer vision and pattern recognition, 2018, pp. 6857–6866.
- [209] Y. Zhang, X. Shi, S. Mi, X. Yang, Image captioning with transformer and knowledge graph, *Pattern Recognition Letters* 143 (2021) 43–49.
- [210] H. Xu, C. Jiang, X. Liang, L. Lin, Z. Li, Reasoning-rcnn: Unifying adaptive global reasoning into large-scale object detection, in: Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition, 2019, pp. 6419–6428.
- [211] A. Tiwari, S. Varma, Learning semantic image attributes using image recognition and knowledge graph embeddings, *arXiv preprint arXiv:2009.05812* (2020).
- [212] M. Palmonari, P. Minervini, Knowledge graph embeddings and explainable ai, *Knowledge Graphs for Explainable Artificial Intelligence: Foundations, Applications and Challenges*, IOS Press, Amsterdam (2020) 49–72.
- [213] A. Seeliger, M. Pfaff, H. Krcmar, Semantic web technologies for explainable machine learning models: A literature review., *International Semantic Web Conference* 2465 (2019) 1–16.
- [214] J. Wang, J. Liu, W. Bi, X. Liu, K. He, R. Xu, M. Yang, Improving knowledge-aware dialogue generation via knowledge base question answering, in: Proceedings of the AAAI Conference on Artificial Intelligence, volume 34, 2020, pp. 9169–9176.
- [215] X. Huang, J. Zhang, D. Li, P. Li, Knowledge graph embedding based question answering, in: Proceedings of the Twelfth ACM International Conference on Web Search and Data Mining, 2019, pp. 105–113.
- [216] Y. Zhang, H. Dai, Z. Kozareva, A. Smola, L. Song, Variational reasoning for question answering with knowledge graph, in: Proceedings of the AAAI Conference on Artificial Intelligence, volume 32, 2018.

Declaration of interests

☒ The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

☐ The authors declare the following financial interests/personal relationships which may be considered as potential competing interests: